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„MASTER OF SCIENCE (M.SC.)“

Text-Guided Detection of Focal Cortical Dysplasia Type II from MR Images

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Declaration of Authorship

I declare that the work presented here is original and the result of my own investigations. Formulations and ideas taken from other sources are cited as such. It has not been submitted, either in part or whole, for a degree at this or any other university. During the preparation of the text, I used AI-based language models exclusively for language-related assistance, such as rephrasing, grammatical correction, stylistic refinement, and conceptual clarification of code-related issues. The scientific content, methodological decisions, and conclusions are entirely my own.

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Abstract

Numerous methods have been developed for tumor detection in medical images, demonstrating the effectiveness of deep learning across a wide range of diagnostic tasks. However, the detection of Focal Cortical Dysplasia (FCD) is considerably more challenging. In contrast to tumors, these lesions do not exhibit progressive growth, often lack clear visual boundaries, and are characterized by subtle and heterogeneous imaging patterns. As a result, the sensitivity of existing automated detection methods remains relatively low, even for state-of-the-art approaches such as MELD, which currently represents one of the best-performing frameworks in this domain. In addition, publicly available annotated datasets for FCD are highly limited.

To address these limitations, this thesis proposes a novel multimodal segmentation framework that combines surface-based visual features with text-based inputs that guide the segmentation process. Experimental results showed that the proposed approach improves both detection accuracy and sensitivity compared to the MELD baseline. To further illustrate the practical utility of the proposed method, a web-based interface was implemented, enabling access to all model variants and their corresponding predictions.

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1 Introduction

This chapter first highlights the importance of epilepsy detection and discusses the limitations of existing methods. It outlines the main objectives of the thesis and its research contributions. Finally, the chapter concludes with an overview of the thesis structure.

1.1 Motivation

Automatic detection of tumors using medical imaging has been widely studied across different internal organs. Recent advances in deep learning and transformer-based approaches have led to impressive results in tumor detection tasks, including lung cancer [1], brain tumors [2], kidney tumors on Computed Tomography (CT) imaging [3], and breast cancer [4]. These studies demonstrate the potential of modern computer vision techniques to achieve clinically significant results in various diagnostic tasks.

However, the detection of epileptogenic lesions, in particular FCD, remains substantially more challenging.

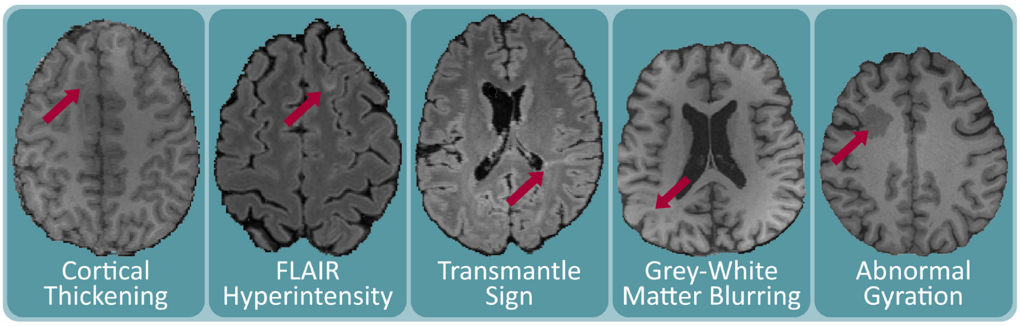


Figure 1.1: Typical features of FCDs visible in Magnetic Resonance Imaging (MRI) [5]

Unlike tumors, which often show progressive growth and well–defined boundaries, epileptogenic lesions typically remain stable in size and present with subtle imaging features, as illustrated in Figure 1.1, complicating their identification even for expert neuroradiologists [5]. Earlier approaches [6–8] to automated FCD detection mainly used volumetric convolutional neural networks trained on 3D Magnetic Resonance Imaging (MRI) data, such as T1-weighted and FLAIR images. However, these methods were usually developed using relatively small datasets, often including only a few dozen to a few hundred patients, which limited their reliability and ability to generalize to unseen data. In addition, volumetric CNN-based approaches do not directly model the cortical surface or its shape, even though FCD-related abnormalities mainly affect the cortex. In contrast, the Multicentre Epilepsy Lesion Detection (MELD) project [9] uses a surface-based representation of the cortex and applies graph neural networks to model cortical structure and local relationships, allowing the use of large multi-center datasets and addressing important limitations of earlier volumetric methods. Although this represents one of the most advanced solutions to date, performance remains limited, underscoring the continued complexity of the detection task.

At the same time, recent progress in text-guided and multimodal learning suggests that combining visual information with textual descriptions may help models focus on clinically relevant anatomical regions. In the context of epileptogenic lesion detection, such textual information can reflect prior clinical knowledge derived from examinations other than MRI, such as Electroencephalography (EEG) findings or seizure semiology. For tasks with limited training data, textual annotations can therefore serve as an additional source of structure by guiding the model toward regions that are clinically suspected, even when MRI findings are subtle. This motivates investigating whether integrating such information, even in a coarse form (e.g., hemisphere or lobe labels), can help compensate for the scarcity of annotated datasets and improve detection performance.

In this thesis, we build upon the surface-based feature maps produced by the pretrained MELD model, which represent the cortical surface as a sparse graph. Given this graph-based representation, a natural next step is to investigate whether additional Graph Neural Network (GNN) layers can further improve lesion detection by aggregating information across larger neighborhoods on the cortical surface. Understanding the benefits and limitations of such graph-based aggregation is particularly important in sparse settings, where capturing long-range context may directly influence lesion detection.

1.2 Aims of the Thesis

The main objective of this thesis is to investigate how the integration of textual and visual information can improve the accuracy of epileptogenic lesion detection. Several research questions were formulated:

1. Do incorporating textual descriptions from anatomical atlases influence the accuracy of lesion segmentation?
2. Can competitive performance be achieved using only partial atlas information (e.g., hemisphere or lobe labels)?
3. What is the impact of integrating additional GNN blocks with MELD-based representations on segmentation quality?

We systematically evaluate the impact of different types of textual information, as well as alternative strategies for combining visual features, including the integration of additional GNN layers. We further analyze how these design choices influence segmentation performance, demonstrating that the joint integration of textual and visual features can enhance both the accuracy and the sensitivity of epileptogenic lesion detection.

1.3 Contributions

The main contributions of this thesis can be summarized as follows:

1. Introduced a state-of-the-art multimodal model for epileptogenic lesion detection that integrates surface-based visual features with textual information.

2. Conducted a systematic evaluation of design choices related to graph-based feature aggregation, including different configurations of GNN blocks.

3. Assessed the effect of multiple types of textual descriptions (full atlas annotations, hemisphere and lobe labels, etc.) on model accuracy, sensitivity and specificity.

4. Delivered comprehensive documentation and usage guidelines for the implemented methods.

5. Implemented a user-friendly interface that supports different pretrained models and enables interactive visualization of predictions.

1.4 Thesis structure

The structure of this thesis is organized as follows:

1. Chapter 2 introduces the core architectural paradigms used in this thesis: transformer architecture, graph neural networks, and large language models, and subsequently reviews state-of-the-art methods for FCD detection, including volumetric CNN-based models, surface-based neural network approaches, and a multimodal segmentation model.

2. Chapter 3 presents the proposed multimodal segmentation framework for epileptogenic lesion detection, detailing the surface-based visual features (from MELD preprocessing), the GNN block, LM textual embeddings, and their integration within the GuideDecoder, and describes the loss function and evaluation metrics.

3. Chapter 4 describes the Bonn and MELD datasets, FreeSurfer-based feature extraction and harmonization, the data-augmentation pipeline, and the variants of atlas-derived textual descriptions used in our study.

4. Chapter 5 describes the main implementation and training settings of the proposed model, outlines the hardware configuration used for the experiments, and presents the web interface developed to facilitate interaction with different model variants.

5. Chapter 6 details the experimental setup and reports the main results. It includes baseline comparisons with MELD, an analysis of the effect of linking different numbers of MELD feature stages to the GNN block, and experiments with mixed and incorrect textual descriptions.

6. Chapter 7 discusses the experimental findings, emphasizing trade-offs between sensitivity and precision, limitations of the current study, implications for clinical applicability, and directions for future research.

7. Chapter 8 summarizes the main contributions and key findings of this thesis.

2 Related Works

This chapter firstly reviews the transformer architecture, which is a fundamental component of modern vision models and large language models (LLMs). Then it introduces graph neural networks (GNNs) and LLMs, followed by an overview of existing approaches to FCD detection and text-guided medical image segmentation for tumor detection tasks, with a focus on their fusion strategies, and limitations.

2.1 Transformer Architecture

The transformer architecture, introduced in the seminal work by Vaswani et al. [10], is based entirely on attention mechanisms and does not rely on recurrence or convolution. A standard Transformer consists of an encoder and a decoder, each composed of stacked layers combining attention mechanisms and position-wise feed-forward networks (see Figure 2.1).

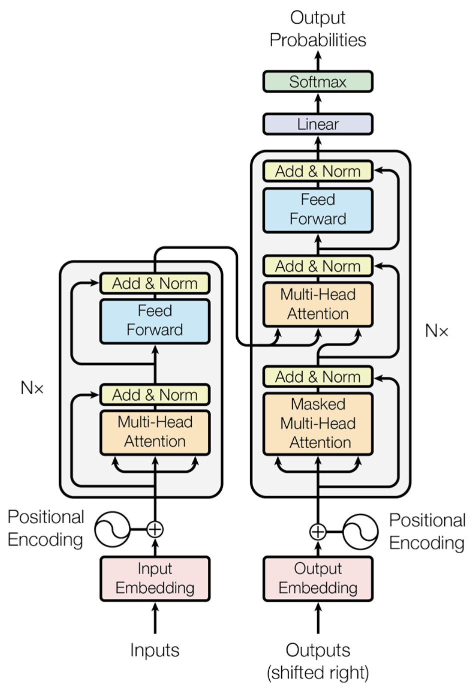


Figure 2.1: The Transformer – model architecture [10].

Let $X \in \mathbb{R}^{N \times d_{\text{model}}}$ denote the input sequence embeddings, where N is the sequence length and d_{model} is the embedding dimensionality. Since the Transformer has no inherent notion of order, positional encodings are added to X to inject information about token positions. A commonly used fixed sinusoidal positional encoding is defined as

$$\mathrm{PE}(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right), \tag{2.1}$$

$$\mathrm{PE}(pos, 2i+1) = \cos\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right), \tag{2.2}$$

where pos denotes the position in the sequence and i indexes the embedding dimensions. The positional encodings have the same dimensionality as X and are combined with the input embeddings by element-wise addition.

Each encoder layer contains a multi-head self-attention module followed by a position-wise feed-forward network, with residual connections and layer normalization applied after each sub-layer. The core operation is scaled dot-product self-attention (see Figure 2.2), defined as

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{2.3}$$

where d_k is the dimensionality of the key vectors.

In self-attention, the query, key, and value matrices are obtained by linear projections of the same input X:

$$Q = XW^Q, \quad K = XW^K, \quad V = XW^V, \tag{2.4}$$

with learnable projection matrices $W^Q, W^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$ and $W^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$.

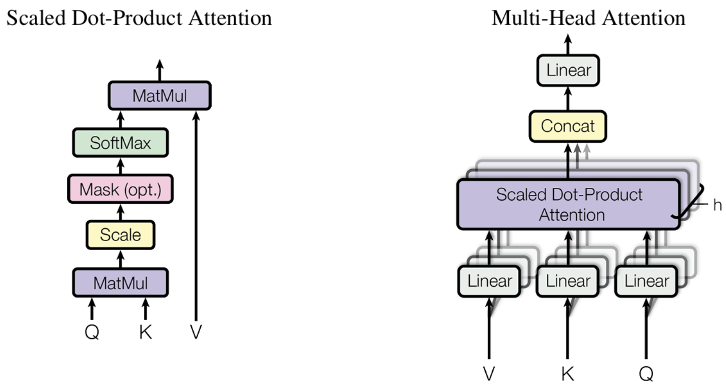


Figure 2.2: (left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel [10].

To increase representational capacity, Transformers employ multi-head attention (see Figure 2.2), which performs h attention operations in parallel:

$$\mathrm{MultiHead}(Q, K, V) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)\, W^O, \tag{2.5}$$

where

$$\mathrm{head}_i = \mathrm{Attention}(XW_i^Q, XW_i^K, XW_i^V), \tag{2.6}$$

and $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$ projects the concatenated outputs back to the model dimension.

Each attention block is followed by a position-wise feed-forward network (FFN), applied independently to each sequence element:

$$\mathrm{FFN}(x) = \sigma(xW_1 + b_1)W_2 + b_2, \tag{2.7}$$

where $\sigma(\cdot)$ denotes a non-linear activation function. In the original Transformer formulation, $\sigma(\cdot)$ is the ReLU activation.

In addition to self-attention, Transformers support *cross-attention*, which enables information exchange between two different modalities or sequences. In cross-attention, the queries are derived from one input X_q, while keys and values are obtained from another input X_k:

$$Q = X_q W^Q, \quad K = X_k W^K, \quad V = X_k W^V. \tag{2.8}$$

Originally developed for natural language processing, Transformer–based models have since been successfully adapted to computer vision (e.g., Vision Transformers and hierarchical variants) and to multimodal learning scenarios. In text-guided medical image segmentation, Transformers are commonly used either as text encoders or as fusion modules, where cross-attention integrates semantic information from textual descriptions with spatial image representations.

2.2 Graph Neural Networks

Graph Neural Networks (GNNs) [11] provide a principled framework for learning on non-Euclidean data such as meshes, surfaces, and general irregular structures. In contrast to convolutional architectures that operate on grids, GNNs propagate information along graph edges, enabling feature integration across anatomically or structurally meaningful neighborhoods. Most modern GNNs follow the *message-passing* formulation [12], where node representations are iteratively updated by aggregating features from adjacent nodes. For a node v with neighborhood $\mathcal{N}(v)$ and current features h_v, the generic message-passing layer is expressed as:

$$m_v = \mathrm{AGG}\big(\{\, \phi(h_v, h_u, e_{uv}) : u \in \mathcal{N}(v) \,\}\big), \tag{2.9}$$

$$h'_v = \gamma(h_v, m_v), \tag{2.10}$$

where ϕ computes messages from neighbors, AGG is a permutation-invariant aggregation function (e.g., mean, sum, max), and γ is an update function (often implemented as an MLP). Different GNN architectures instantiate these components in distinct ways. For example, Graph Convolutional Networks (GCN) [13] use a normalized mean aggregator, while GraphSAGE [14] generalizes aggregation through mean, pooling, or LSTM-based operators, enabling inductive learning on unseen graphs.

Hierarchical GNN architectures extend the message-passing paradigm to multi-resolution settings, analogous to U-Net models [15] in Euclidean domains and graph U-Net architectures [16]. They downsample and upsample graphs using pooling and unpooling operators, which enable coarse-to-fine feature extraction on complex surfaces. Unlike grid-based pooling in CNNs, graph pooling operates on irregular structures and therefore requires an explicit strategy for reducing the number of nodes while preserving important topological and semantic information.

In pooling layers, a coarser graph is constructed by selecting or aggregating nodes from the original graph, thereby reducing the graph resolution. This can be achieved through different mechanisms, such as node clustering, score-based node selection, or predefined hierarchical mappings. Formally, pooling maps node features $H \in \mathbb{R}^{N \times d}$ to a reduced representation $H' \in \mathbb{R}^{N' \times d'}$ with $N' < N$, while also defining a corresponding coarser adjacency structure.

Unpooling operations perform the inverse transformation by projecting features from the coarse graph back to a finer resolution. This is commonly implemented using stored assignment indices or interpolation operators, allowing features from coarse nodes to be distributed to their corresponding fine-scale nodes. Skip connections between pooling and unpooling stages are often employed, analogous to U-Net architectures, to preserve high-resolution information and stabilize gradient flow.

Such hierarchical pooling–unpooling schemes enable GNNs to capture both local geometric details and global contextual information, which is particularly important for cortical surface analysis, where anatomical structures are naturally represented at multiple spatial scales.

Building on these principles, the MELD project [9] applies a GNN-U-Net architecture to multi-resolution cortical graphs and demonstrates that graph-based convolutions are effective for detecting subtle morphological abnormalities associated with FCD.

In this thesis, these principles are leveraged to integrate cortical surface features with textual clinical information, enabling multimodal segmentation within a unified GNN-based framework.

2.3 Large Language Models

Large Language Models (LLMs) are typically based on transformer architectures and are trained on large text corpora to learn contextual semantic representations.

Given an input text sequence, the LLM first applies a tokenizer that converts the raw text into a sequence of discrete tokens. Each token is represented by an integer index from a fixed vocabulary $\mathcal{V}$, defined as the set of all tokens supported by the tokenizer. These indices are mapped to continuous vectors via a learned embedding matrix $E \in \mathbb{R}^{|\mathcal{V}|\times d}$, where $|\mathcal{V}|$ denotes the vocabulary size and d the embedding dimensionality. The embedding matrix E is a trainable parameter of the model and is learned jointly with the remaining LLM weights during pretraining through gradient-based optimization.

For a token with index i, the corresponding token embedding is obtained as

$$x_i = E[i], \tag{2.11}$$

i.e., by selecting the i-th row of the embedding matrix. This yields a sequence of token embeddings $\mathbf{X} = (x_1, \ldots, x_T)$.

The LLM encoder subsequently maps these token embeddings to contextualized representations by means of stacked self-attention layers. At each transformer layer, token embeddings are projected into query, key, and value vectors, and interactions between all token pairs are computed via scaled dot-product attention. This mechanism allows each token to attend to all other tokens in the sequence and to integrate information from the global context. After multiple transformer layers, the encoder produces a sequence of hidden states

$$H = (h_1, \ldots, h_T), \tag{2.12}$$

where each vector h_i represents the final context-dependent embedding of token x_i, jointly encoding its semantic content and its relationships to surrounding tokens.

LLMs are typically trained using one of two objectives. In *causal* language modeling (used in GPT-style models), the model predicts the next token based on the previously observed ones:

$$p(x_{i+1} \mid x_1, \ldots, x_i) = \mathrm{softmax}(W^O h_i). \tag{2.13}$$

In *masked* language modeling (MLM), employed by BERT-style encoders, randomly selected tokens are masked and reconstructed from the surrounding context:

$$p(x_i \mid X_{\setminus i}) = \mathrm{softmax}(W^O h_i), \tag{2.14}$$

where $X_{\setminus i}$ denotes the input sequence with token x_i masked.

LLMs trained with these objectives produce rich, high-level semantic embeddings that capture lexical, syntactic, and domain-specific patterns in text [17]. In medical imaging, domain-adapted LLMs such as BioBERT [18] provide contextual text embeddings that can guide segmentation models through cross-attention or contrastive alignment [19], improving robustness and interpretability in multimodal medical applications.

2.4 State-of-the-Art in FCD Detection

Building on early vision-based approaches in medical imaging, initial automated methods for FCD detection relied on convolutional neural networks operating directly on MRI data. Dev et al. [7] proposed a fully convolutional neural network with an encoder–decoder structure and skip connections, conceptually similar to a U-Net, for the automatic detection and localization of FCD from volumetric MRI. Their approach performs voxel–wise segmentation on multimodal MRI data, enabling direct lesion identification without manual feature engineering. However, the method was trained on a relatively small, single–center dataset, which may limit its robustness and generalizability across different populations and imaging protocols.

Similarly, House et al. [8] introduced a clinically integrated 3D convolutional neural network for the automated detection and segmentation of FCD lesions from routine T1–weighted and FLAIR MRI sequences. The proposed architecture employs a shared encoder that extracts latent representations from the MRI data, followed by two decoders: one dedicated to FCD segmentation and a variational decoder tasked with reconstructing the input to model normal brain anatomy. This joint training strategy regularizes the encoder and improves segmentation performance, particularly in settings with limited labeled data. A key contribution of this work is its prospective validation within a real–world clinical workflow. Nevertheless, the model exhibits substantially lower detection performance for subtle or non-sclerotic FCD lesions compared to more conspicuous cases, suggesting a bias toward visually pronounced lesion patterns and limited generalization across the full pathological spectrum.

Despite their architectural differences, both approaches operate purely in volumetric image space and rely primarily on intensity-based MRI features. As a result, they remain sensitive to noise, imaging artifacts, and inter-scanner variability, and do not explicitly model cortical morphometric characteristics such as cortical thickness, curvature, or gray–white matter blurring, which are key markers of FCD. These limitations make subtle lesions particularly difficult to detect and motivate the use of surface-based representations that better capture cortical anatomy.

A more recent and substantial advancement is represented by the MELD Graph model [9], which formulates FCD detection as a surface-based segmentation problem on the cerebral cortex. Starting from cortical surface reconstructions obtained with standard neuroimaging pipelines (e.g., FreeSurfer [20]), the cortex is discretized into a mesh, where vertices correspond to cortical surface points and edges encode local neighborhood relationships. Each vertex is associated with a set of morphometric and intensity-based attributes, including cortical thickness, curvature, gray–white matter contrast, and other surface-derived features known to be relevant for FCD characterization.

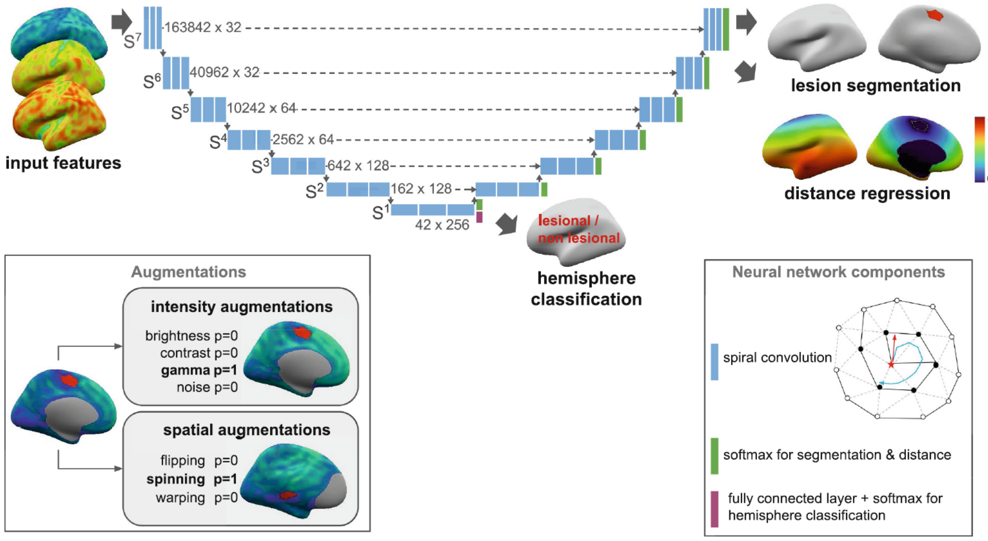


Figure 2.3: Architecture of MELD: Proposed GC-nnU-Net model for lesion segmentation, with auxiliary distance regression and hemisphere classification tasks [9].

These surface meshes are represented as multi-resolution graphs, enabling the application of a GNN-U-Net architecture (see Figure 2.3) that performs hierarchical feature aggregation and pooling directly on the cortical surface. Trained on the largest multi-center FCD dataset available to date, the MELD Graph model represents the current state of the art in surface-based FCD detection and demonstrates strong robustness to inter-site variability between imaging centers. Owing to its surface-based representation of cortical anatomy and its strong empirical performance, the MELD Graph model serves as a well-established vision backbone for the present study.

Recent progress in multimodal learning suggests that incorporating textual information alongside visual features can improve performance in data–limited medical imaging tasks. In particular, Ariadne's Thread [21] demonstrated that integrating textual descriptions via a cross–attention mechanism leads to consistent performance gains for tumor detection, even when only a small number of annotated training samples is available. This finding is especially relevant for FCD detection, where annotated data are scarce and lesion appearance is highly heterogeneous. Motivated by these results, we investigate whether similar benefits can be achieved by augmenting a strong surface-based vision model with textual anatomical information.

3 Method

This chapter presents the proposed method, including the model architecture, its individual components, as well as the loss functions and evaluation metrics used for training.

As mentioned earlier, the design of the proposed architecture is inspired by the work of [21]. Our implementation differs in two key aspects:

- the depth of the feature extraction pathway is increased, and
- a GNN-based block is introduced to more effectively aggregate visual features at higher resolutions.

An overview of the overall architecture is shown in Figure 3.1, and each component is described in detail below.

3.1 Visual Feature Extraction

As input to the vision model, structural MRI scans are first mapped into the FreeSurfer surface space. From the MELD preprocessing pipeline, we obtain a multi-resolution set of surface-based features across seven hierarchical levels. Each level is represented as a tensor of shape (H, N, C), where:

- H – number of hemispheres, with $H = 2$ corresponding to the left and right hemispheres,
- N – number of vertices on the cortical mesh per hemisphere,
- C – number of features per vertex.

To aggregate higher-order geometric information, we process the selected feature layers individually through a dedicated **GNN Block**. This design choice is based on empirical findings (see Chapter 6), where incorporating the lowest-resolution layers led to oversmoothing and degradation of discriminative information, while using only the higher-resolution layers provided the best trade-off between expressivity and stability.

3.2 GNN Block

Formally, let $\mathbf{X}^{(l)} \in \mathbb{R}^{(B \cdot H \cdot N_l) \times C_l}$ denote the input feature matrix at layer l, where B is the batch size, H is the number of hemispheres, N_l is the number of vertices per hemisphere at layer l, and C_l is the number of input channels.

Each *GNN Block* then applies the following sequence of operations:

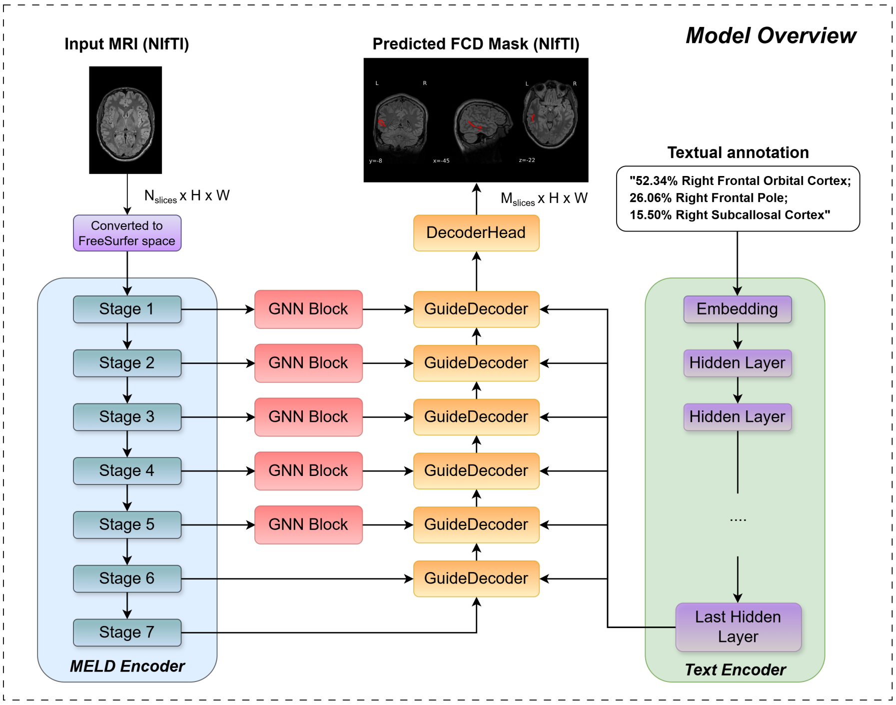


Figure 3.1: Overview of the proposed multimodal segmentation architecture. The input to the MELD encoder is an MRI scan in NIfTI format, which is first transformed into FreeSurfer space, a standardized surface-based cortical representation obtained by reconstructing the cerebral cortex from volumetric MRI. In this space, surface-based visual features include cortical thickness, curvature, sulcal depth, as well as intensity-based measures derived from T1-weighted and FLAIR images. The resulting surface-based visual features from each stage of the MELD encoder are then processed by their corresponding GNN block. Concurrently, atlas-derived textual anatomical annotations are encoded by a pretrained language model, and the visual and textual representations are fused through the GuideDecoder blocks. Each stage has its own GuideDecoder, which integrates the multimodal information and passes it to the next level via upsampling. Finally, the fused features are decoded into the predicted FCD mask, which is output in NIfTI format. *Note.* The number of GNN blocks shown is illustrative and does not reflect a fixed architectural requirement; it is included solely for visualization purposes.

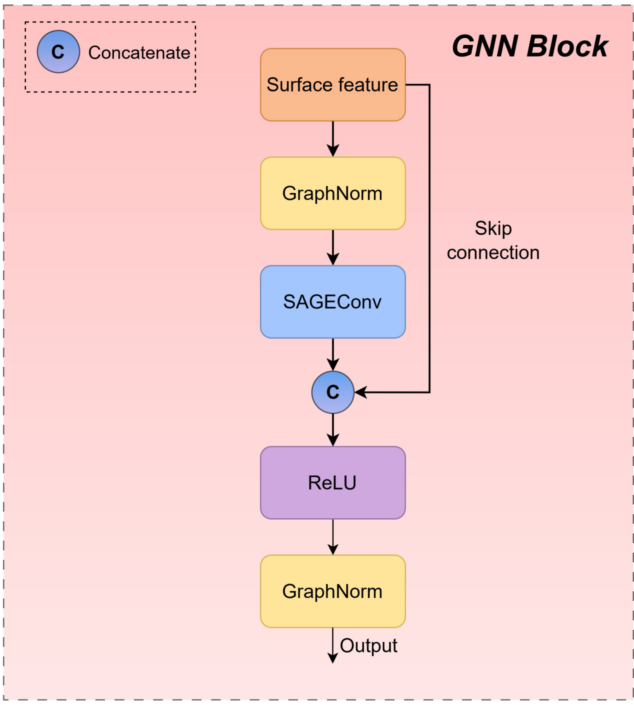


Figure 3.2: Structure of the GNN Block used in the MELD encoder. Each surface-based feature is normalized with GraphNorm, processed through a SAGEConv layer, and concatenated with the original feature via a skip connection. The result is passed through ReLU activation and another GraphNorm layer to produce the output representation.

$$\mathbf{H}_0 = \mathrm{GraphNorm}\big(\mathbf{X}^{(l)}, \texttt{batch}\big)\,, \tag{3.1}$$

$$\mathbf{H}_1 = \mathrm{SAGEConv}(\mathbf{H}_0, \texttt{edge\_index}_l)\,, \tag{3.2}$$

$$\mathbf{H}_2 = \mathbf{H}_1 + \mathbf{H}_0 \quad \text{(residual connection)}, \tag{3.3}$$

$$\mathbf{H}_3 = \mathrm{ReLU}(\mathbf{H}_2)\,, \tag{3.4}$$

$$\mathbf{Z}^{(l)} = \mathrm{GraphNorm}(\mathbf{H}_3, \texttt{batch})\,. \tag{3.5}$$

where:

- $\texttt{edge\_index}_l \in \mathbb{N}^{2\times|E_l|}$ — adjacency structure of the cortical surface graph;
- batch — batch vector;
- $\mathbf{Z}^{(l)} \in \mathbb{R}^{(B\cdot H\cdot N_l)\times C_l}$ — resulting representation for level l;
- **SAGEConv** implements the GraphSAGE [14] update rule

  $$\mathbf{x}'_i = \mathbf{W}_1\mathbf{x}_i + \mathbf{W}_2 \cdot \underset{j\in\mathcal{N}(i)}{\mathrm{mean}}\,\mathbf{x}_j,$$

  where $\mathcal{N}(i)$ is the neighborhood of node i;
- **GraphNorm**$(x) = \gamma \cdot \dfrac{x-\mu_g}{\sqrt{\sigma_g^2+\epsilon}} + \beta$, where μ_g, σ_g^2 are mean and variance per graph, and γ, β are learnable parameters;

- **ReLU**$(x) = \max(0, x)$ — rectified linear activation function;

The choice of **GraphNorm** is motivated by its ability to normalize node features on a per-graph basis. In practice, individual graphs can vary substantially in size, topology, and feature distribution, and normalizing across an entire batch may therefore yield unstable or inconsistent statistics. By computing normalization statistics separately for each graph, GraphNorm preserves the overall feature distribution within that graph and provides more stable activations, which is particularly important when training on heterogeneous cortical graphs. In addition, we adopt the pretrained **SAGEConv** operator as the main aggregation mechanism, since it is widely used and has proven effective for combining information from neighboring nodes in graph-based models.

After passing through the GNN Block, the feature dimensionality remains unchanged, ensuring consistency with subsequent blocks. The resulting features are then passed to the GuideDecoder for multimodal fusion.

3.3 Textual Feature Extraction

The text branch of our architecture leverages the pretrained **RadBERT** model [22], which was selected as an initial choice due to its specialization in radiology reports. In Chapter 6, we additionally evaluate alternative pretrained text encoders to assess whether domain adaptation or model size affects segmentation performance.

The output of RadBERT's final hidden layer is passed to each GuideDecoder to enable multimodal fusion with cortical graph features.

Although previous studies suggest that selectively unfreezing the last transformer layers may improve downstream performance [23], we do not explore this direction in the present work and leave it for future investigations.

3.4 GuideDecoder

We adopt the GuideDecoder architecture (Figure 3.3) proposed by Zhong et al. [21], which fuses visual and textual features in a multimodal manner. The decoder first receives the text embeddings produced by the pretrained language model and projects them into the visual feature space so that their dimensionality matches that of the image tokens. Positional encodings are then added to both modalities to preserve their ordering and support the attention mechanisms operating within the decoder.

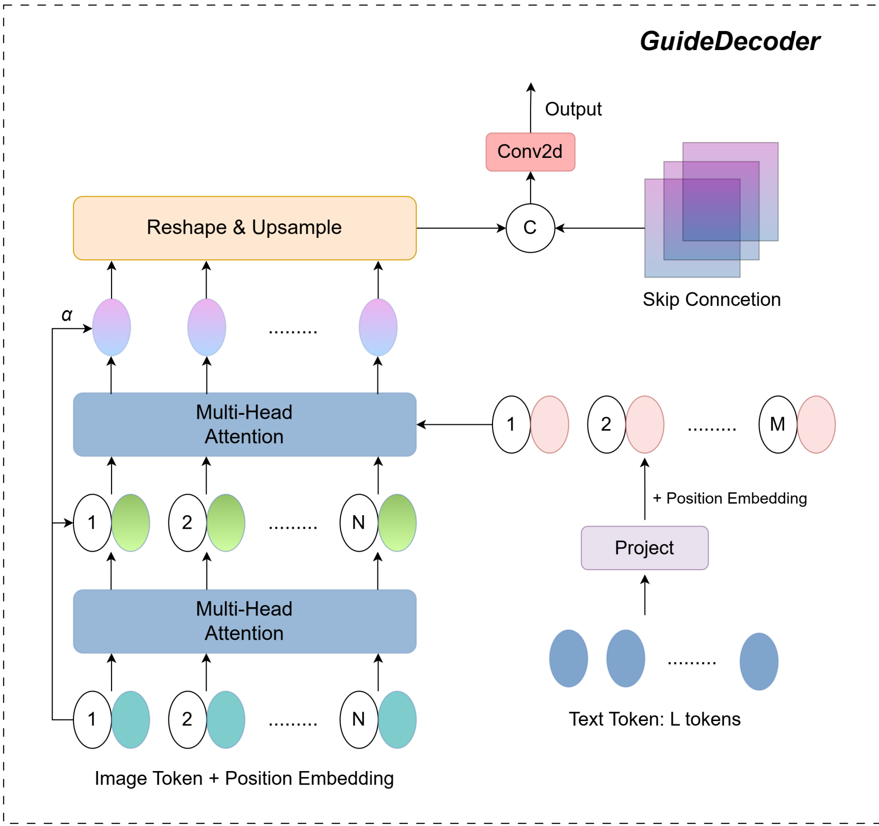


Figure 3.3: Structure of the GuideDecoder block. At each stage, the block receives image tokens from the visual encoder and text embeddings projected into the visual feature space. The visual tokens are refined through multi-head attention and fused with the projected text tokens via cross-attention. The fused multimodal features are then upsampled and combined with the skip connection before the final convolution produces the stage-specific output.

The visual tokens are processed through a stack of *multi-head attention layers*, which iteratively refine their representations before interacting with the text stream. The projected text tokens serve as keys and values in a *cross-attention module*, allowing the decoder to incorporate semantically relevant information from the textual description into the visual features.

The resulting multimodal representations are subsequently reshaped and upsampled to the spatial resolution required by the decoder stage and fused with the corresponding skip connections from the visual encoder. A final convolutional layer transforms the fused features into the output prediction map.

In our implementation, the overall design of the GuideDecoder is preserved, but we introduce two modifications to adapt the architecture to surface-based representations. First, we replace the standard 2D upsampling with the `HexUnpool` operator (see Appendix HexUnpool), which performs mean unpooling on the icosphere mesh. Second, instead of the original `DynUNetBlock`[1] from the MONAI framework, which consists of convolution, normalization, and activation layers, the standard convolution is substituted with a `SpiralConv` operator [24], enabling convolutional feature aggregation directly on the cortical surface mesh after upsampling and fusion with skip connections.

[1] https://docs.monai.io/en/0.3.0/_modules/monai/networks/blocks/dynunet_block.html

3.5 Loss function

Following best practices for medical image segmentation and to ensure a fair comparison with the MELD model, we employed a composite loss. The loss function is defined as

$$L = L_{ce} + L_{dice} + L_{dist} + L_{class} + \sum_{i \in I_{ds}} w_{ds}^{i} \cdot (L_{ce}^{i} + L_{dice}^{i} + L_{dist}^{i}) \tag{3.6}$$

The individual components are detailed below.

Cross-Entropy Loss

For the segmentation task, we employ the binary cross-entropy loss, which is commonly used in medical image segmentation. Here, y_i denotes the ground-truth label, $\hat{y}_i$ the predicted probability, and n the number of vertices:

$$L_{ce} = -\frac{1}{n} \sum_{i=1}^{n} \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right]. \tag{3.7}$$

Dice Loss

The Dice Loss L_{dice} directly optimizes for the overlap between predicted and ground-truth lesion masks. This loss is less sensitive to class imbalance and encourages the network to predict coherent lesion regions. It is defined as

$$L_{dice} = 1 - \frac{2 \sum_{i=1}^{n} y_i \hat{y}_i}{\sum_{i=1}^{n} y_i^2 + \sum_{i=1}^{n} \hat{y}_i^2 + \epsilon} \tag{3.8}$$

It is important to note that we considered two different implementations of this loss. In the MONAI library, the Dice score is computed for each sample and then averaged over the batch (macro–averaging), whereas in the MELD implementation the aggregation is performed immediately over the entire batch (micro–averaging). In our experiments, we observed that under strong class imbalance between background and lesion voxels, the micro–averaged version shifts the loss contribution towards the background, which reduces the gradient signal for rare lesion voxels and leads to training instability. Based on these empirical observations, we used the MONAI implementation, which provided higher metric values and enabled the model to detect more FCDs.

Distance Loss

To provide the network with additional contextual information and reduce false positives, we include a distance regression loss L_{dist}. The model is trained to predict the normalized geodesic distance d_i from each vertex to the lesion boundary. Here, $\hat{y}_{i,0}$ denotes the model's output corresponding to the *non-lesional* class for vertex i. We employ a mean absolute error loss weighted by $(d_i + 1)^{-1}$, which reduces the contribution of distant non-lesional vertices to the loss. Since the weight decreases with distance from the lesion boundary, the network focuses more on regions located closer to the lesion [9]:

$$L_{dist} = \frac{1}{n} \sum_{i=1}^{n} \frac{|d_i - \hat{y}_{i,0}|}{d_i + 1}. \tag{3.9}$$

Classification Loss

In the final configuration, we add a weakly supervised classification head to mitigate the uncertainty between lesion masks and actual lesions. Each subject is labeled as positive if any of its vertices belongs to a lesion. The classification head aggregates features across all vertices at the deepest level (level 1), producing a single subject-level prediction $\hat{c}$. The classification loss is computed at the subject level and summed over a mini-batch of n samples [9].

$$L_{class} = -\sum_{i=1}^{n} c_i \log(\hat{c}_i) + (1 - c_i) \log(1 - \hat{c}_i). \tag{3.10}$$

Deep Supervision

To encourage gradient flow and stabilize training, we adopt deep supervision at intermediate decoder levels $I_{ds} = \{6, 5, 4, 3, 2, 1\}$. At each level i, auxiliary predictions are generated and the same combination of cross-entropy, dice, and distance losses is applied. These auxiliary losses are weighted by w_{ds}^i and added to the total objective. The deep supervision weights w_{ds}^i are fixed hyperparameters adopted from the MELD framework, where their values are explicitly specified [9].

$$\sum_{i \in I_{ds}} w_{ds}^i (L_{ce}^i + L_{dice}^i + L_{dist}^i). \tag{3.11}$$

3.6 Metrics

To evaluate the performance of our model, we follow the metrics used by the MELD authors for direct comparison: Dice score, Positive Predictive Value (PPV), Intersection over Union (IoU), specificity, and sensitivity. For consistency, we also use the same strategy as the MELD authors to compute confidence intervals. Below, we briefly describe each of them.

Dice score

The Dice similarity coefficient (DSC) is defined as the harmonic mean between precision and recall. For two sets of predicted positives P and ground truth positives G, it is given by

$$\text{Dice}(P, G) = \frac{2\,|P \cap G|}{|P| + |G|} = \frac{2TP}{2TP + FP + FN}, \tag{3.12}$$

where TP, FP and FN denote true positives, false positives and false negatives, respectively.

Positive Predictive Value

Also known as precision, PPV measures the proportion of correctly identified positive samples among all predicted positives:

$$\text{PPV} = \frac{TP}{TP + FP}. \tag{3.13}$$

In this work, we report PPV at two granularities:

- **Pixel-level PPV** evaluates precision over individual vertices of the cortical surface and therefore reflects how accurately the predicted mask matches the ground-truth lesion at a fine spatial scale.
- **Cluster-level PPV** treats each connected predicted region as a single detection and counts it as a true positive only if it overlaps with the manual lesion mask (or, following the MELD protocol, lies within 20 mm of it).

Intersection over Union

The IoU (also called the Jaccard index) quantifies the overlap between predicted and ground truth labels:

$$\text{IoU}(P, G) = \frac{|P \cap G|}{|P \cup G|} = \frac{TP}{TP + FP + FN}. \tag{3.14}$$

Specificity

Specificity quantifies the model's ability to avoid false-positive detections in healthy controls. It is defined as the proportion of control subjects for whom the model predicts no lesion clusters. Let $y_i = 0$ denote control subjects and $\hat{y}_i = 0$ indicate that the model produced no predictions for subject i. Then specificity is given by

$$\text{Specificity} = \frac{\sum_{i=1}^{N} \mathbb{1}(y_i = 0 \wedge \hat{y}_i = 0)}{\sum_{i=1}^{N} \mathbb{1}(y_i = 0)}. \tag{3.15}$$

Higher specificity indicates fewer false alarms in subjects without FCD.

Sensitivity

Sensitivity measures the ability of the model to correctly detect lesions in patients with FCD. Following the MELD evaluation protocol, a prediction is counted as correct if at least one predicted cluster overlaps with the manual lesion mask or lies within a 20 mm distance from it. Formally, for a set of patients with ground-truth labels $y_i \in \{0, 1\}$ indicating the presence of FCD and binary prediction outcomes $\hat{y}_i \in \{0, 1\}$, sensitivity is defined as

$$\text{Sensitivity} = \frac{\sum_{i=1}^{N} \mathbb{1}(y_i = 1 \wedge \hat{y}_i = 1)}{\sum_{i=1}^{N} \mathbb{1}(y_i = 1)}. \tag{3.16}$$

A higher value indicates that the model successfully identifies a larger proportion of true FCD cases.

Confidence Intervals

To quantify the uncertainty of the reported metrics, we compute 95% confidence intervals following the procedure used by the MELD authors. For each metric, we report the sample median together with a confidence interval estimated via a non-parametric percentile bootstrap.

Given scores $x_1, \ldots, x_N$, we draw $B = 10{,}000$ bootstrap resamples of size N with replacement and compute the median for each resample, yielding the bootstrap distribution $\{\tilde{x}^{(b)}\}_{b=1}^{B}$. The 2.5th and 97.5th percentiles of this distribution are taken as the lower and upper bounds of the 95% interval.

This method makes no assumptions about the underlying distribution; missing values (NaN) are removed prior to resampling. When $N = 1$, the interval collapses to the observed value. A fixed random seed is used for reproducibility.

4 Dataset

We gratefully acknowledge the **MELD Project** for providing access to the dataset used in this work. Without this resource, it would not have been possible to conduct a systematic evaluation of our model.

This chapter describes the datasets used in this work, including preprocessing steps, surface extraction, data augmentation strategies, and text description generation.

4.1 Bonn Dataset

The Bonn scientific group has published a presurgical MRI dataset on **OpenNeuro**, entitled "*An open presurgery MRI dataset of people with epilepsy and focal cortical dysplasia type II*" [25]. This dataset comprises **170 participants**, including **85 individuals with FCD type II** and **85 healthy controls**. For each participant, the following data are available:

- High-resolution **3D T1-weighted** MRI scans (isotropic voxels, $1\,\text{mm}^3$ or $0.8\,\text{mm}^3$ resolution, depending on the subject);
- Corresponding **isotropic 3D FLAIR** imaging;
- **Manually delineated regions of interest (ROIs)** identifying FCD lesions, provided for patients only;
- A set of **clinical and demographic variables** (including age, sex, lesion laterality and location, histopathological subtype IIa/IIb, MRI-negative status, and postoperative outcome according to Engel classification).

Dataset Structure

The dataset follows the BIDS (Brain Imaging Data Structure) convention, with each participant stored in an individual directory. A typical subject folder has the following structure:

```
sub-00001/
    anat/
        sub-00001_XXX-XXXXX_T1w.nii.gz
        sub-00001_XXX-XXXXX_T1w.json
        sub-00001_XXX-XXXXX_FLAIR.nii.gz
        sub-00001_XXX-XXXXX_FLAIR.json
        sub-00001_XXX-XXXXX_FLAIR_roi.nii.gz     (patients only)
```

Each NIfTI image is accompanied by a corresponding JSON sidecar file containing DICOM-derived metadata, including voxel size, TR/TE/TI parameters, image orientation, and scanner information.

At the dataset root, several metadata files are provided:

```
participants.tsv
participants.json
participants_with_scanner.tsv
dataset_description.json
```

The file `participants.tsv` contains demographic and clinical information (e.g., age, sex, diagnosis, lesion laterality, histopathological subtype), while `participants.json` provides descriptions of these variables. The file `participants_with_scanner.tsv` includes additional scanner-related metadata such as field strength and vendor. The standard BIDS file `dataset_description.json` documents the dataset-level metadata.

Manually delineated lesion masks (`*_roi.nii.gz`) are available only for patients and are spatially aligned with the anatomical T1-weighted MRI.

4.2 MELD Dataset

The **MELD Project** provides a large-scale neuroimaging dataset comprising MRI scans and clinical data of patients with FCD, as well as healthy controls. In total, the dataset includes **1185 participants** from **23 international epilepsy surgery centers**, including the Bonn datasets described earlier. Due to missing or corrupted files in the obtained release, we used a subset of **1130 participants** in our experiments. The dataset is not publicly available; access must be requested directly from the study authors [9].

For each participant, the dataset includes:

- Structural MRI: **3D T1-weighted** (acquired for all participants) and **3D FLAIR**, which was included only for participants for whom this sequence was available;
- Lesion annotations: **manually delineated Region of Interest (ROI)** identifying FCD lesions (patients only). For MRI-negative cases, i.e., patients in whom no visible lesion is detected on conventional MRI despite histologically confirmed FCD, **postsurgical resection cavities** were used to guide ROI definition;
- Demographic and clinical metadata;
- Predefined splits for training, validation, and test cohorts.

Dataset Structure

To reduce inter-site variability arising from differences in scanners and acquisition protocols across centers, the MELD dataset provides surface-based features that have been harmonized using the ComBat method [26]. The provided ComBat-harmonized `.hdf5` files do not contain the multiscale representations used in the MELD pipeline. Instead, they store only per-vertex cortical attributes extracted by FreeSurfer (e.g., cortical thickness, curvature, sulcal depth, and T1/FLAIR intensity features), together with their asymmetry maps and the corresponding lesion masks.

Clinical and demographic metadata are distributed separately in CSV/TSV files. The available metadata fields include: *participant ID, scanning site, diagnostic group, age of onset,* *epilepsy duration, age at presurgical MRI, sex, MRI-negative status, Engel outcome, histopathology, lesion hemisphere and lobe, presence of FLAIR hyperintensity, seizure outcome,* and scanner information.

The dataset release also includes predefined train/validation/test splits, which we follow to ensure comparability with prior MELD studies.

Cohorts and evaluation protocol

In this thesis, the MELD dataset is used as the **main cohort**. We follow the predefined MELD train/validation/test split for model development and report results on the corresponding **MELD test set**.

The Bonn dataset is treated as an **independent cohort** and is used **only for testing**.

In this thesis, the MELD dataset is used as the main cohort. We follow the predefined MELD train/validation/test split for model development and report results on the corresponding MELD test set. Although the Bonn dataset is formally included in the MELD collection, it is treated as an independent cohort and used exclusively for testing, following the original MELD evaluation protocol.

4.3 FreeSurfer Processing

Each image was processed with the FreeSurfer framework [20], from which **11 core surface-based features** were extracted:

- **Morphometric features:** cortical thickness, sulcal depth, curvature, and intrinsic curvature;
- **Intensity features:** gray–white matter intensity contrast, and FLAIR intensity sampled at 6 intracortical and subcortical depths.

In addition to the raw measurements, features were further processed into **raw values**, **control-normalized features**, and **asymmetry features** (left vs. right hemisphere). Cortical thickness was additionally adjusted by regressing out curvature. Altogether, this yielded **34 input features per participant**, computed at **163,842 vertices** on a bilaterally symmetrical cortical surface template [9]. Scanner-related variability was addressed through the ComBat harmonization applied in the released MELD features.

For conversion from volume space to FreeSurfer surface space, detailed instructions are available in the MELD documentation [27]. In practice, however, it is often more practical to request preprocessed files directly from the authors, since converting a single image to surface space is computationally expensive: approximately 6–7 hours per scan (even with FastSurfer [28] on an NVIDIA A100 GPU, the process required 3–4 hours).

4.4 Data Augmentation

In the original study MELD authors applied augmentations in three stages:

- **Lesion–mask augmentation**: deforming the lesion region together with its corresponding lesion mask and surface-based features to produce anatomically coherent augmented samples;

- **Mesh–space transforms**: applying geometric transformations to the cortical surface mesh, consistently affecting both features and lesion masks;

- **Intensity transforms**: modifying the intensity features at each vertex.

Each transform is applied independently with probability p defined in the experiment configuration. The order is fixed: lesion–mask augmentation $\rightarrow$ mesh transforms $\rightarrow$ intensity transforms.

Lesion–mask augmentation

Given a geodesic distance map D on the cortical surface (negative inside the lesion), we first normalise it by $|\min D|$ and add low–frequency noise, which is generated on a low–resolution icosphere (level 2) and then upsampled to the target resolution using the predefined unpool operators:

$$\tilde{D} = \texttt{Unpool}\Big(\tfrac{D}{|\min D|} + \mathcal{N}(0,\sigma^2)\Big), \qquad L' = \mathbb{1}\{\tilde{D} \le 0\}.$$

In our implementation, we use $\sigma = 0.5$ by default. And as `Unpool` we use the `HexUnpool` operator (see Appendix HexUnpool). After modifying the binary lesion mask L, the geodesic distances and smoothed labels are **recomputed**:

$$D' = \texttt{fast\_geodesics}(L'), \qquad \widetilde{L} = \texttt{smoothing}(L', \text{iteration} = 10).$$

This procedure is applied only if the lesion mask is non-empty. After augmentation, the geodesic distances and smoothed labels are recomputed to ensure a consistent lesion shape. However, in our training setup, only the resulting binary lesion mask is used for supervision, while the recomputed distances and smoothed labels are not used directly in the loss function.

Mesh–space transforms (icosphere re–indexing)

We use precomputed vertex index mappings provided by the MELD authors, which define fixed permutations of vertices on the icosphere mesh. These mappings are generated offline for each transformation type (spinning, warping, and flipping) and are applied uniformly to all vertex-wise tensors (features, labels, distances, etc.) to preserve anatomical correspondence (GitHub repository). The MELD framework defines three types of such maps:

- **Spinning** — index remapping that corresponds to a rigid rotation of the icosphere.

- **Warping** — smooth non-rigid remapping that locally compresses or stretches the mesh.

- **Flipping** — mirror reflection of the mesh, e.g. exchanging left and right hemispheres.

Per-vertex feature intensity transforms

All per-vertex intensity transformations are applied to all feature channels. For each transformation, parameters are sampled per channel; unless stated otherwise, the same parameters are applied consistently across all vertices within a channel:

- **Gaussian noise**: add $\epsilon \sim \mathcal{N}(0, \sigma^2)$ with $\sigma^2 \sim \mathcal{U}(0, 0.1)$.

- **Brightness scaling**: multiply each channel by $m_c \sim \mathcal{U}(0.75, 1.25)$.
- **Contrast adjustment**: for each channel c, sample $f \sim \mathcal{U}(0.65, 1.5)$ and apply

$$x_c \leftarrow (x_c - \mu_c)\, f + \mu_c,$$

$$min_c \coloneqq \min(x_c), \quad max_c \coloneqq \max(x_c),$$
$$x_c[x_c < min_c] = min_c, \quad x_c[x_c > max_c] = max_c.$$

- **Gamma**: for $\gamma \sim \mathcal{U}(0.7, 1.5)$,

$$x' \leftarrow \Big(\frac{x - \min(x)}{\max(x) - \min(x) + \varepsilon}\Big)^{\gamma} (\max(x) - \min(x) + \varepsilon) + \min(x);$$

$$x'' \leftarrow \frac{x' - \mu}{\sigma + \varepsilon}, \quad \text{where } \varepsilon = 10^{-7},\ \mu = \text{mean}(x),\ \sigma = \text{std}(x).$$

We also use an *inverted* variant by applying the same operation to $-x$ and negating back.

Disabled/placeholder transforms

The current implementation contains placeholders for `Gaussian blur` and `low-resolution downsampling`. These operations are not implemented in the original authors' code and were not used in our reported results.

4.5 Types of Atlas Descriptions

We generated textual descriptions with `AtlasReader` [29]. Given each ROI, it is first registered to template space and anatomical atlases are overlaid to quantify its anatomical context. For every atlas region, we compute the percentage of the ROI volume that overlaps that region and report the resulting region–percentage table. When visual inputs are augmented, the corresponding textual descriptions are regenerated from the transformed lesion location to remain anatomically consistent, ensuring alignment between visual and textual modalities. In this work, we used the Harvard–Oxford (cortical and subcortical) probabilistic atlas [30].

However, strong validation scores obtained using full anatomical descriptions do not guarantee robustness at deployment: in real-world scenarios, neither precise overlap percentages nor complete region names are typically available. Therefore, in Chapter 6, we also evaluate several reduced-text settings, summarized in Table 4.1.

The anatomical names of lobes and their respective regions are taken from *Automated Anatomical Labeling of Activations in SPM Using a Macroscopic Anatomical Parcellation of the MNI MRI Single-Subject Brain*.

Table 4.1: Text conditioning variants.

Name	Description	Example
full	Complete atlas–based lesion description with region names and their percentages	48.86% Left Middle Frontal Gyrus; 30.63% Left Precentral Gyrus; ...
hemi	Hemisphere only	Left hemisphere
lobe_regions	Lobe regions only (subset of atlas regions)	Middle Frontal Gyrus; Temporal Pole; ...
hemi+lobe_regions	Hemisphere + lobe regions	Left hemisphere; Middle Frontal Gyrus; Temporal Pole ...
hemi+lobe	Hemisphere + lobe name	Left hemisphere; Frontal lobe; Temporal lobe; Limbic lobe
no_text	No textual conditioning is provided (equivalent to assuming the full brain without additional localisation cues)	full brain
mixed	randomly sampling one of {hemisphere only, lobe region only, hemisphere + lobe, full text, no text}; this scheme is described in more detail in the next chapter	–

Note: For healthy subjects (no lesion present), the textual input is set to *"No lesion detected."*

5 Implementation

This chapter describes the main implementation settings used for training the proposed model. It outlines the key characteristics of the hardware on which the experiments were conducted and presents the web interface developed to facilitate interaction with the different model variants.

5.1 Training Settings

In this work, we aim to follow the training protocol of the MELD authors to ensure a fair and consistent comparison. The proposed framework was trained using the following hyperparameters.

The training batch size was fixed at 8 and the validation batch size at 4. The initial learning rate was set to 3×10^{-4} and optimized using the OneCycleLR scheduler (maximum learning rate 3×10^{-3}). The schedule included a warm-up phase during the first 10% of training steps, followed by cosine annealing. Training was performed for up to 100 epochs (minimum 20 epochs), with early stopping applied if the validation performance did not improve for 20 epochs. For evaluation, we trained an ensemble of five independently initialized models (seeds 42–46). At test time, we averaged their per-vertex predictions to form the final output.

The model integrated surface-based graph features and contextual text embeddings. Feature channel dimensions across the encoder stages were $[32, 32, 64, 64, 128, 128, 256]$, with corresponding text sequence lengths $[128, 64, 64, 32, 32, 16, 16]$, and a maximum text length of 256 tokens. Deep supervision was employed with levels $I_{ds} = [6, 5, 4, 3, 2, 1]$ and associated weights $w_{ds} = [0.5, 0.25, 0.125, 0.0625, 0.03125, 0.0150765]$ [9]. The text encoder was initialized from RadBERT, with a projection dimension of 768.

To address class imbalances, non-lesional hemispheres were undersampled, ensuring that approximately one third of training examples contained a lesion. Data augmentation strategies included:

Table 5.1: Data augmentation strategies applied during training [9].

Transformation	**Probability**
Random flipping	0.5
Gaussian blur	0.2
Spinning	0.2
Warping	0.2
Brightness/contrast/gamma	0.15 each
Gaussian noise	0.15
Low-resolution simulation	0.25

Hardware

All experiments were performed on the *Bender* high-performance computing cluster at the University of Bonn, equipped with NVIDIA A100 GPUs (80 GiB each) and AMD EPYC CPUs.

A detailed description of the system can be found in the official documentation[1]. Our implementation used PyTorch 2.1.0+cu121, TorchVision 0.16.0+cu121, TorchAudio 2.1.0+cu121, Torch Geometric 2.5.3, Torch Scatter 2.1.2, TorchMetrics 0.11.4, and Python 3.9.18.

To ensure reproducibility, a complete software environment has been provided via Docker. The corresponding Dockerfiles and installation instructions are available in our GitHub repository[2].

5.2 Web Interface

To facilitate the use of different pretrained models, we developed a web interface that provides an accessible and intuitive way to interact with the lesion detection system.

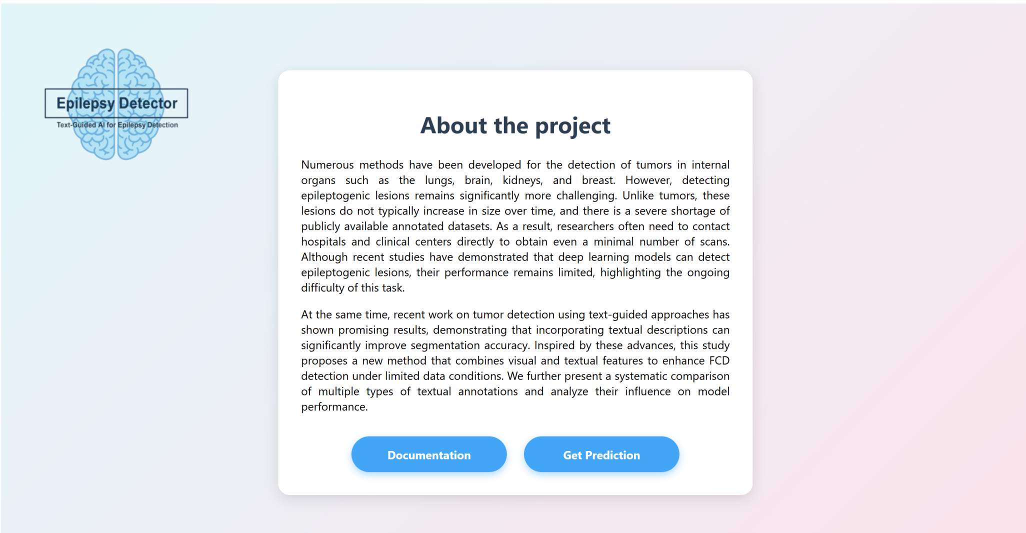


Figure 5.1: Main page of the web interface with a brief project description and navigation buttons.

The main page presents a brief overview of the project and includes navigation controls that let users explore the documentation or move on to the prediction page for processing a specific patient (see Fig. 5.1).

[1]https://www.hpc.uni-bonn.de/en/systems/bender
[2]https://github.com/Wektor607/FCD-Detection/tree/docker-inference-fix/meld_graph

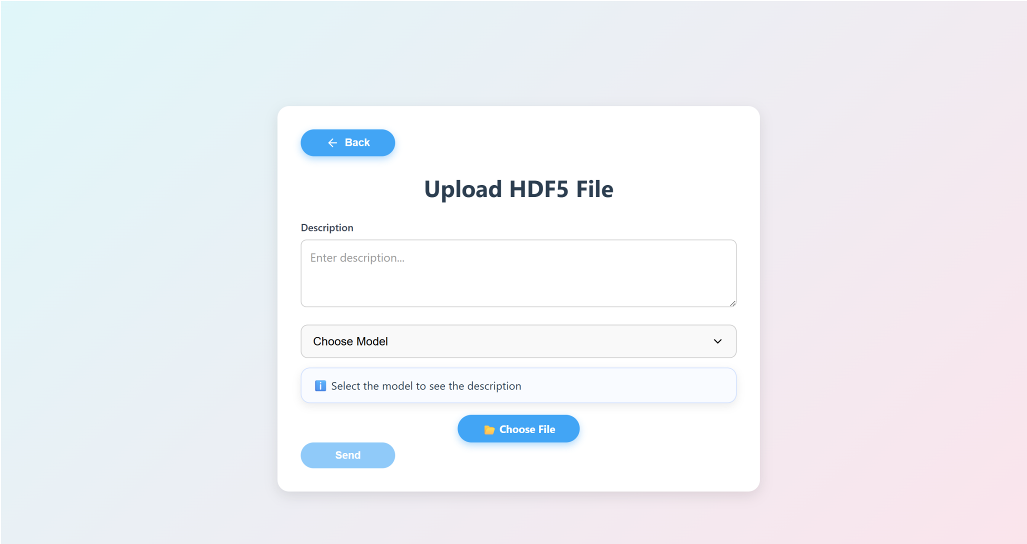


Figure 5.2: Upload page where the user selects a model, adds a description, and uploads an HDF5 file for analysis. *Note: the current prototype accepts only a single HDF5 file.*

The workflow begins on the upload page, where the user selects one of the trained models, optionally provides a short textual description, and uploads an HDF5 file for analysis (see Fig. 5.2).

For multimodal models, the description field is mapped to a textual input channel. If the field is left empty, a default neutral description (e.g., "full brain") is used, representing the absence of explicit anatomical guidance. This allows comparing informative and non-informative textual conditions without changing the model architecture. Vision-only models ignore the description field entirely. The current prototype supports the upload of a single HDF5 file at a time.

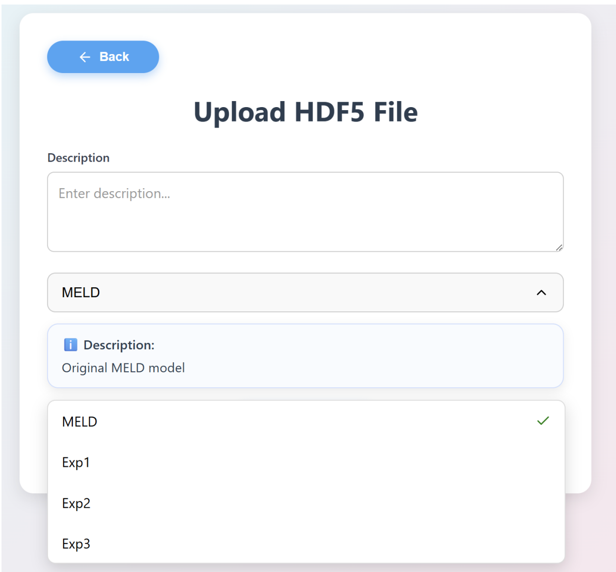


Figure 5.3: Dropdown menu for selecting among trained models.

To support different use cases and experimental settings, the interface includes a model selection menu listing all available trained variants. These include the baseline MELD model, two vision-only architectures (`Vision-only` and `Vision-only+SA`) with minor architectural differences, and the multimodal `Vision+LM` model trained on various textual inputs. Each option is accompanied by a short explanation, similar in style to modern LM model pickers, helping users choose between faster or more accurate configurations (see Fig. 5.3).

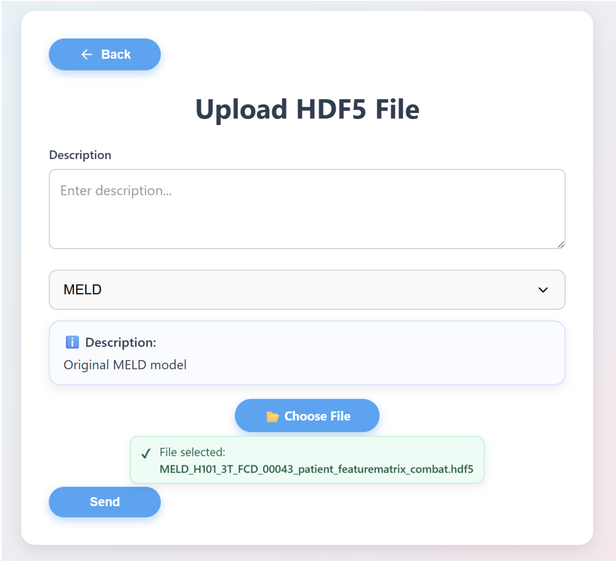


Figure 5.4: Example of a successfully uploaded file ready for processing.

Once a file is uploaded successfully, the interface confirms that the case is ready for processing and allows the user to initiate inference (see Fig. 5.4). After the prediction is complete, the results page provides two complementary views of the output:

(i) representative 2D MRI slices with the predicted lesion regions overlaid for quick inspection, and

(ii) a 3D cortical surface visualization rendered for both hemispheres (left and right), shown from two complementary viewpoints (anterior and posterior) to improve interpretability and spatial understanding.

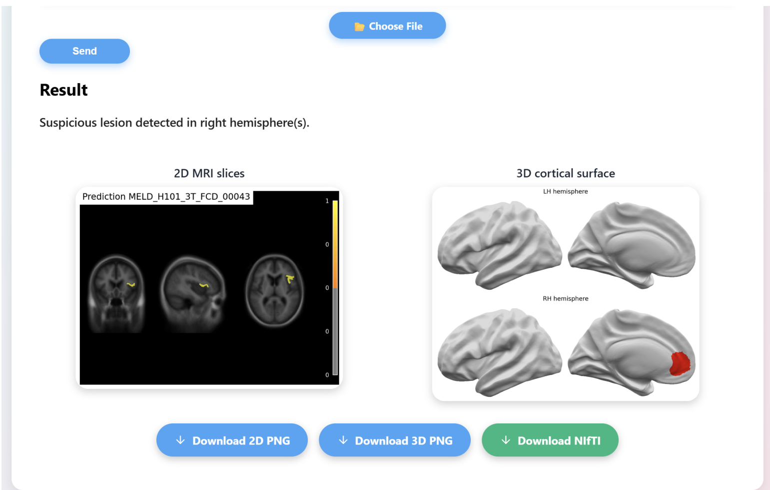
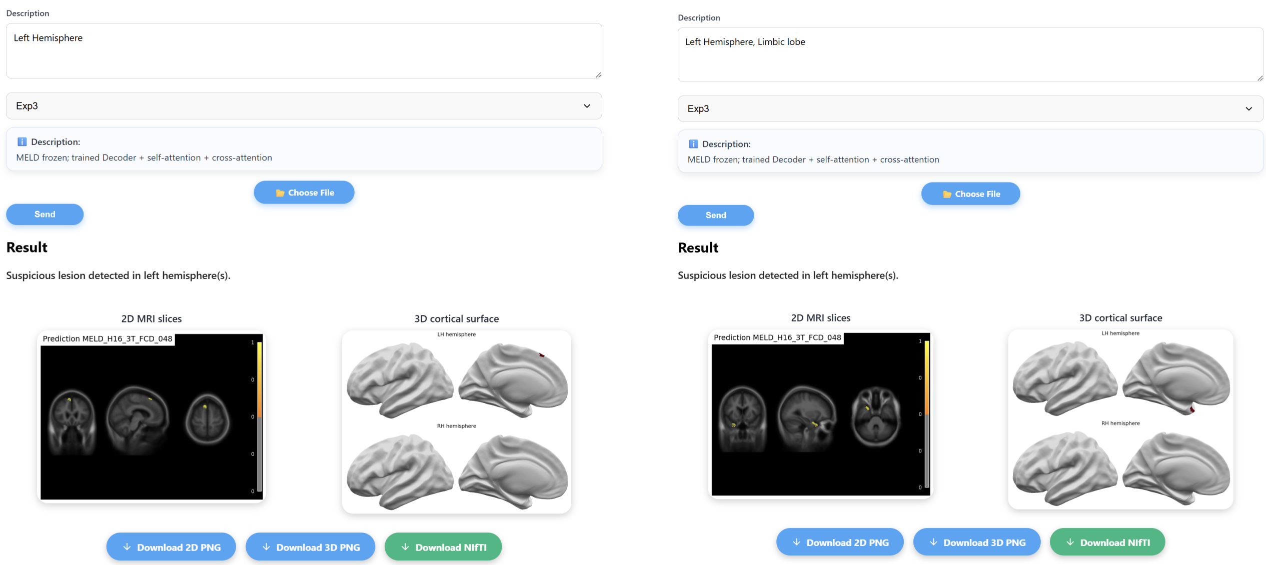


Figure 5.5: Web interface showing the inference output: 2D MRI slice overlays and a 3D cortical surface rendering for both hemispheres (LH/RH) from complementary viewpoints (anterior/posterior). All outputs can be exported as PNG (2D/3D) and as a NIfTI volume for further analysis in 3D medical imaging tools.

All generated visualizations can be downloaded as PNG files (2D slices and 3D renderings), and the full volumetric prediction is additionally provided in NIfTI format for further analysis in standard 3D medical imaging software (see Fig. 5.5). For the 2D view, the interface displays a fixed, pre-rendered image showing three representative MRI slice views with overlaid predictions, intended for qualitative illustration. The 2D output is not interactive, and slice selection or navigation is not supported in the current prototype.

This design supports both general users (rapid visual validation) and developers/researchers (downstream processing and quantitative evaluation).

To demonstrate the influence of textual input on the final prediction, we selected a single patient and applied two different text descriptions: (i) *"Left hemisphere"* and (ii) *"Left hemisphere, Limbic lobe"*. Despite using identical imaging data, the resulting predictions differ notably. This comparison demonstrates that different textual descriptions, while using identical imaging data, can lead to noticeably different prediction outcomes (see Fig. 5.6).



(a) Text input: "Left hemisphere" **(b)** Text input: "Left hemisphere, Limbic lobe"

Figure 5.6: Comparison of model predictions for the same patient using different textual descriptions. More anatomically specific input leads to better localization of the predicted lesion.

Implementation Details From a technical perspective, the interface follows a client–server design. The frontend is implemented in React and provides model selection, data upload, and visualization components. The backend is implemented using FastAPI and handles file validation, preprocessing, and model inference. Communication between frontend and backend is performed via REST endpoints, with inference results returned as volumetric predictions and pre-rendered visualizations.

6 Experiments

This chapter presents the main experimental results, focusing on the influence of different textual prompts on model performance. It includes experiments with mixed and incorrect textual descriptions, as well as an analysis of how the addition of GNN blocks affects the final segmentation results and the challenges encountered in this setting.

6.1 Basic Experiments

To systematically assess the contribution of each component of the proposed multimodal architecture, we gradually enable individual decoder modules— geometry-based upsampling, self-attention, and finally text-conditioned cross-attention. This stepwise experimental design allows isolating the effect of textual guidance and ensures that any improvement in lesion detection quality can be attributed to a specific architectural modification rather than confounding changes.

In all configurations, we freeze both encoders: the visual encoder (the **MELD** backbone) and the text encoder. Although **RadBERT** was used as the initial language model during the early stages of this work—primarily because it was the first domain-specific model that integrated smoothly into the pipeline—the choice of text encoder was later examined more systematically. After observing that textual signals substantially influence lesion detection quality, we conducted an additional study comparing several biomedical and radiology-oriented language models.

These language models differ in the domain and scale of their pretraining data: **RadBERT** and **BlueBERT** are pretrained on radiology reports and clinical text, while **PubMedBERT** is trained on large-scale biomedical literature. The results of this analysis (see Appendix Semantic similarity analysis) motivated the inclusion of multiple text-encoder variants in the experimental evaluation.

Across all variants, the geometry-based upsampling path (`HexUnpool` + `SpiralConv`) and the final segmentation head are always trained, while both encoders remain frozen (Figure 6.3). The experimental variants differ only in (i) whether the `GuideDecoder` is inserted before the upsampling path and (ii) the type of atlas-derived textual input used. The pretrained `Vision-only` model (described below) is used to initialize the decoder for all variants except the MELD-only baseline, which is trained from scratch.

We consider the following configurations:

- **MELD.** The pretrained MELD model was evaluated on the test set without any additional training or fine-tuning and is used as the baseline (see Figure 2.3).

- **Vision–only (Unpool + Spiral, no text).** In this configuration, the pretrained MELD model is used to provide multi–scale visual features extracted from its decoder at each stage. The text branch and the self–/cross–attention modules of the `GuideDecoder` are disabled, while the geometry–based upsampling components are retained (see Figure 6.1). The decoder features from MELD are reshaped and passed directly through the upsampling path (`HexUnpool` + `SpiralConv`) and then to the segmentation head. No textual input is used in this variant, so the prediction relies solely on image–derived features.

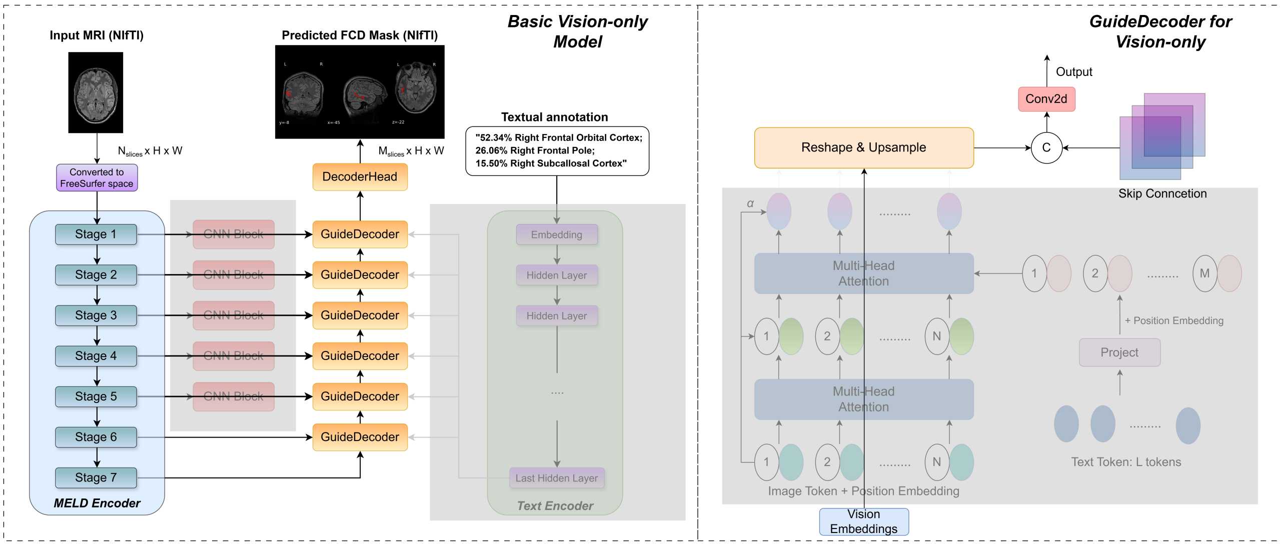


Figure 6.1: Architecture of Vision–only: the text branch and the self-/cross-attention modules in the `GuideDecoder` are disabled, and MELD vision features are fed directly into the geometry-based upsampling path and segmentation head.

- **Vision–only+SA (GuideDecoder: self-attention only).** As in `Vision-only`, multi–scale visual features are extracted from the pretrained MELD decoder and used as input to the `GuideDecoder`. One `GuideDecoder` block is inserted at each decoder level (D6–D1), i.e., one block per upsampling stage, to refine the features using self–attention before the subsequent upsampling step (see Figure 6.2). The text branch remains disabled; therefore, cross–attention is not applied and each `GuideDecoder` performs self–attention only. The upsampling path and segmentation head are kept identical to `Vision-only`.

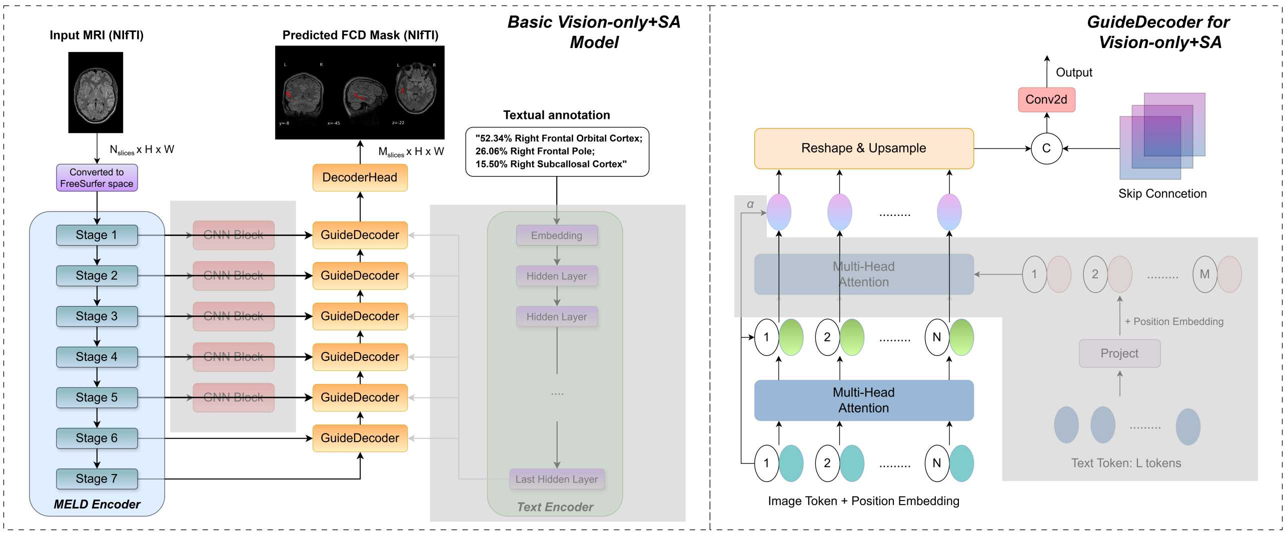


Figure 6.2: Architecture of `Vision-only+SA`: each `GuideDecoder` block contains a self-attention head, while the text branch remains disabled.

- **Vision+LM (GuideDecoder + Text).** As in `Vision-only` and `Vision-only+SA`, multi–scale visual features are extracted from the pretrained MELD decoder and used as input to the `GuideDecoder`. In contrast to `Vision-only+SA`, the full `GuideDecoder` is enabled, i.e., self–attention on image tokens and cross–attention between image and text embeddings are applied (see Figure 6.3). The only difference across variants is the textual conditioning (Table 4.1):

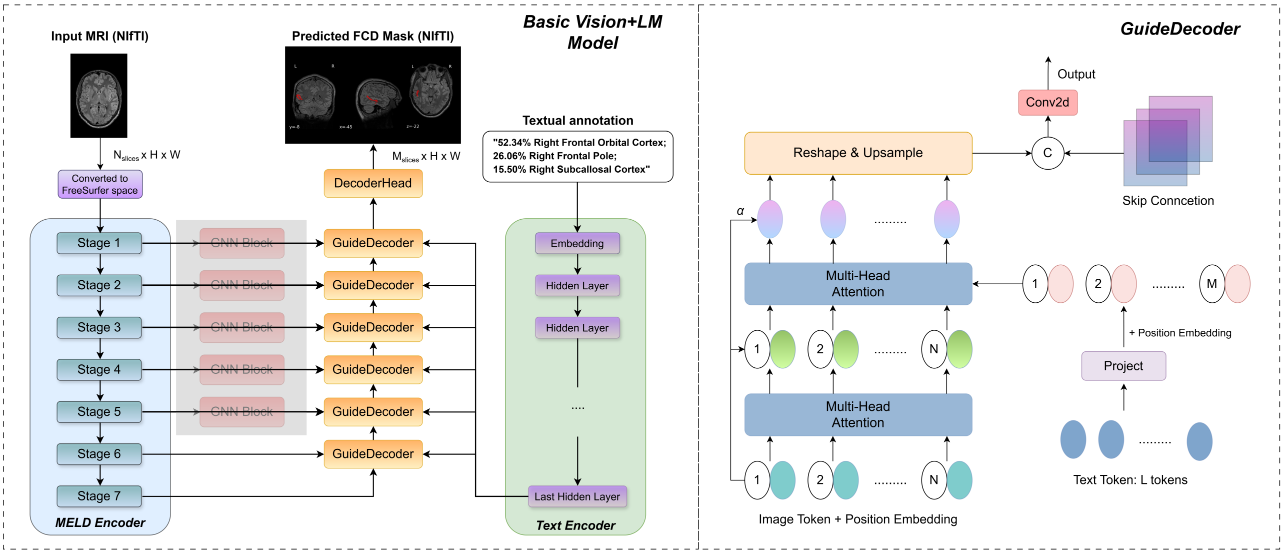


Figure 6.3: Architecture of `Vision+LM`: the full `GuideDecoder` is used, incorporating both self-attention on visual tokens and cross-attention between visual features and text embeddings.

- **Vision+LM_mixed (GuideDecoder + Mixed Text).** Same as **Vision+LM**, but Atlas descriptions were randomly sampled from the text conditioning variants listed in Table 4.1.

  For each subject, a full Atlas–based description was generated once and all partial textual variants were derived from it. During training, one textual variant was randomly sampled independently for each sample at every data loading step. Consequently, the same subject could be paired with different textual inputs across epochs.

Main cohort

Table 6.1 presents the median performance on the MELD test set (main cohort). In the RadBERT group, defined as all variants using the default RadBERT language encoder (without alternative LMs such as BlueBERT or PubMedBERT), the prompt combining *hemisphere and lobe-region* terms detected the most lesions, though with only slightly higher Dice, $\text{PPV}_{\text{pixels}}$, and IoU. Across all models, the PubMedBERT variant achieved the highest number of detected lesions, likely due to better domain-context understanding (see Table 1 in the Appendix). This clearly demonstrates that selecting an appropriate language model can materially improve performance, underscoring the substantial impact of the language encoder.

In real-world settings, detailed subregion information may be unavailable. We therefore evaluated prompts with only general *lobe* names (e.g., *Frontal lobe*, *Temporal lobe*, *Insular lobe*). For the RadBERT-based models, performance was very close to the *hemi + lobe_regions* setup: Dice decreased by ∼2.5 percentage points (p.p.), PPV_pixels by ∼0.4 p.p., IoU by ∼1.6

Table 6.1: Median performance on the main cohort (with 95% confidence intervals).

Model	Dice	PPV pixels	(mean) PPV clusters	(median) PPV clusters	IoU	Specificity, % (N = 193)	Sensitivity, % (N = 259)
MELD	0.230 (0.107–0.320)	0.166 (0.086–0.220)	0.515	1.000 (1.000-1.000)	0.130 (0.057–0.190)	58.0 (n=112) (51.3–64.8)	65.6 (n=170) (59.8–71.4)
Vision–only	0.238 (0.125–0.313)	0.198 (0.130–0.318)	0.361	1.000 (0.500-1.000)	0.135 (0.066–0.186)	31.1 (n=60) (24.9–37.8)	69.1 (n=179) (63.3–74.9)
Vision–only+SA	0.256 (0.125–0.326)	0.204 (0.122–0.322)	0.333	1.000 (0.500-1.000)	0.147 (0.067–0.195)	24.9 (n=48) (19.2–31.1)	70.7 (n=183) (65.3–76.1)
Vision+LM: hemi	0.231 (0.148–0.285)	0.188 (0.131–0.298)	0.483	0.667 (0.500-1.000)	0.131 (0.080–0.166)	100.0 (n=193) (100.0–100.0)	71.0 (n=184) (65.6–76.4)
Vision+LM: lobe_regions	0.254 (0.159–0.332)	0.231 (0.168–0.320)	0.404	0.667 (0.500-1.000)	0.145 (0.086–0.199)	100.0 (n=193) (100.0–100.0)	72.6 (n=188) (67.2–78.0)
Vision+LM: hemi + lobe_regions	0.264 (0.180–0.342)	0.238 (0.160–0.327)	0.333	0.500 (0.500-1.000)	0.152 (0.099–0.206)	99.5 (n=192) (98.4–100.0)	74.5 (n=193) (69.1–79.9)
Vision+LM: hemi + lobe	0.239 (0.113–0.302)	0.242 (0.166–0.353)	0.456	0.667 (0.500-1.000)	0.136 (0.060–0.178)	99.5 (n=192) (98.4–100.0)	71.4 (n=185) (66.0–76.8)
Vision+LM: hemi + lobe + BlueBERT	0.230 (0.125–0.305)	0.231 (0.146–0.330)	0.443	0.667 (0.500-1.000)	0.130 (0.067–0.180)	99.0 (n=191) (97.4–100.0)	71.8 (n=186) (66.4–77.2)
Vision+LM: hemi + lobe + PubmedBERT	0.233 (0.139–0.330)	0.242 (0.153–0.342)	0.334	0.500 (0.500-0.667)	0.132 (0.074–0.198)	100.0 (n=193) (100.0–100.0)	76.4 (n=198) (71.4–81.5)
Vision+LM: full_desc	0.246 (0.145–0.330)	0.259 (0.178–0.360)	0.478	0.667 (0.500-1.000)	0.141 (0.078–0.197)	94.3 (n=182) (90.7–97.4)	72.6 (n=188) (67.2–78.0)

Note. Colors denote rank within each column: best , second , third .
Metrics: Dice, IoU, PPV_{pixels}, and $PPV_{clusters}$ are computed on patients only (N = 259). Sensitivity is reported for patients (N = 259), whereas specificity is reported for healthy controls (N = 193).
Vision+LM prompt variants: `hemi` – hemisphere prompt; `lobe_regions` – lobe–region prompt; `hemi+lobe` – hemisphere + coarse lobe prompt; `hemi+lobe+BlueBERT` – same as previous but with BlueBERT instead of RadBERT; `hemi+lobe_regions` – hemisphere + lobe–region prompt; `full_desc` – full atlas prompt.
Vision–only and **Vision–only+SA** are ablation baselines.

p.p., and only **8** fewer lesions were detected. Likewise, using only *hemi* descriptions yielded results close to *hemi + lobe*, indicating that even a minimal textual prompt can help identify additional lesion regions.

Also, when using the full textual description (e.g., "52.34% Right Frontal Orbital Cortex; 26.06% Right Frontal Pole; 15.50% Right Subcallosal Cortex"), the model performed worse than `hemi+lobe_regions`. One possible reason is that the full description contains additional numeric details that may be less informative for the segmentation task, for instance due to overlapping region assignments (the same lesion area contributing to multiple atlas regions). Since RadBERT was trained on radiology reports, it may not benefit from such highly structured percentage–based inputs. For comparison, we also tested PubMedBERT, which was trained on a larger biomedical corpus. Although its Dice, PPV and IoU scores were slightly lower than those of the `hemi+lobe` configuration with RadBERT, it nevertheless detected **13 additional lesion clusters**.

We also report specificity, defined as the proportion of healthy controls with no predicted lesion clusters. Across all *Vision+LM models*, specificity was higher than that of MELD by ∼36–42 p.p.. This indicates that the text–conditioned models distinguish healthy from non–healthy cases substantially better.

Throughout this chapter, we do not focus on the median $PPV_{clusters}$, which is the primary metric reported in the original MELD study. As shown in Table 6.1, the median $PPV_{clusters}$ attains high values across many configurations (often reaching 0.667, with overlapping confidence intervals), providing limited discrimination between models. In contrast, the mean $PPV_{clusters}$ exhibits more variation across configurations and is therefore more informative for comparing the methods in our setting. MELD achieves the highest mean $PPV_{clusters}$ on the main cohort, with a small margin of ∼2–3 p.p. over the next two models and a much larger gap to the remaining ones. On the independent cohort (Table 6.2), however, the situation is reversed: the proposed architecture shows improvements of 2–8 p.p. compared with the two best-performing baselines. Overall, the mean $PPV_{clusters}$ reflects model differences more reliably in our experiments. For these reasons, in the following experiments we report only the mean $PPV_{clusters}$, as it allows us to distinguish between models in this study, whereas the median version does not.

Independent cohort

Table 6.2: Median performance on the independent cohort (with 95% confidence intervals).

Model	Dice	PPV pixels	(mean) PPV clusters	(median) PPV clusters	IoU	Specificity, % (N = 83)	Sensitivity, % (N = 82)
MELD	0.358 (0.209–0.465)	0.261 (0.142–0.428)	0.464	1.000 (1.000–1.000)	0.218 (0.117–0.303)	55.4 (n=46) (44.6–66.3)	69.5 (n=57) (59.8–79.3)
Vision–only	0.376 (0.238–0.529)	0.443 (0.243–0.619)	0.350	1.000 (1.000–1.000)	0.232 (0.138–0.359)	39.8 (n=33) (28.9–50.6)	73.2 (n=60) (63.4–82.9)
Vision–only+SA	0.342 (0.155–0.494)	0.381 (0.177–0.651)	0.379	1.000 (0.667–1.000)	0.206 (0.084–0.328)	37.3 (n=31) (27.7–48.2)	69.5 (n=57) (59.8–79.3)
Vision+LM: hemi	0.372 (0.261–0.515)	0.380 (0.181–0.601)	0.468	1.000 (1.000–1.000)	0.228 (0.150–0.347)	100.0 (n=83) (100.0–100.0)	74.4 (n=61) (64.6–84.1)
Vision+LM: lobe_regions	0.373 (0.283–0.507)	0.407 (0.233–0.589)	0.257	1.000 (0.667–1.000)	0.229 (0.165–0.340)	100.0 (n=83) (100.0–100.0)	78.0 (n=64) (68.3–86.6)
Vision+LM: hemi+ lobe_regions	0.363 (0.190–0.513)	0.434 (0.209–0.657)	0.359	1.000 (0.500–1.000)	0.222 (0.105–0.345)	97.6 (n=81) (94.0–100.0)	74.4 (n=61) (64.6–84.1)
Vision+LM: hemi+lobe	0.327 (0.164–0.512)	0.411 (0.236–0.667)	0.545	1.000 (0.667–1.000)	0.196 (0.090–0.344)	98.8 (n=82) (96.4–100.0)	72.0 (n=59) (62.2–81.7)
Vision+LM: hemi+lobe+ BlueBERT	0.362 (0.260–0.498)	0.469 (0.267–0.620)	0.356	1.000 (0.750–1.000)	0.221 (0.150–0.331)	96.4 (n=80) (91.6–100.0)	76.8 (n=63) (67.1–85.4)
Vision+LM: hemi+lobe+ PubmedBERT	0.347 (0.195–0.529)	0.457 (0.202–0.730)	0.258	1.000 (0.500–1.000)	0.210 (0.109–0.360)	100.0 (n=83) (100.0–100.0)	80.5 (n=66) (72.0–89.0)
Vision+LM: full_desc	0.423 (0.267–0.506)	0.484 (0.315-0.693)	0.460	1.000 (1.000–1.000)	0.268 (0.154–0.339)	90.4 (n=75) (83.1–96.4)	73.2 (n=60) (63.4–82.9)

Note. Colors denote rank within each column: best, second, third.
Metrics: Dice, IoU, PPV_{pixels}, and $PPV_{clusters}$ are computed on patients only (N = 82). Sensitivity is reported for patients (N = 82), whereas specificity is reported for healthy controls (N = 83).
Vision+LM prompt variants: `hemi` – hemisphere prompt; `lobe_regions` – lobe–region prompt; `hemi+lobe` – hemisphere + coarse lobe prompt; `hemi+lobe+BlueBERT` – same as previous but with BlueBERT instead of RadBERT; `hemi+lobe_regions` – hemisphere + lobe–region prompt; `full_desc` – full atlas prompt.
Vision–only and **Vision–only+SA** are ablation baselines.

Following the MELD paper, the Bonn cohort is used as an independent test set to assess out-of-distribution generalization.

The models that detected the largest number of lesion clusters were those trained with *lobe_regions* prompts in the RadBERT group, as well as the *hemi + lobe + PubMedBERT* variant across all architectures. Within the *hemi + lobe* configuration, replacing BlueBERT with PubMedBERT led to increased lesion detection sensitivity (80.5% vs. 76.8%). This gain, however, was accompanied by slightly lower segmentation quality compared to the BlueBERT-based variant, as reflected by small decreases in Dice (–1.5 p.p.), PPV_{pixels} (–1.2 p.p.), and IoU (–1.1 p.p.). This trade-off is consistent with PubMedBERT's pretraining on a substantially larger corpus of biomedical and brain-related clinical text, which enhances its ability to recognize lesion-related terminology but does not necessarily improve spatial localization. Overall, the *hemi + lobe + PubMedBERT* model achieved the highest sensitivity (66/82), a key objective in this study, while maintaining acceptable segmentation quality.

Examples

Below, we present two representative cases in which the proposed model produces improved predictions. Each figure shows (from left to right) the ground-truth lesion mask, the prediction of the baseline MELD model, and the prediction of the proposed `Vision+LM` model[1] with PubMedBERT using the `hemi+lobe` prompt.

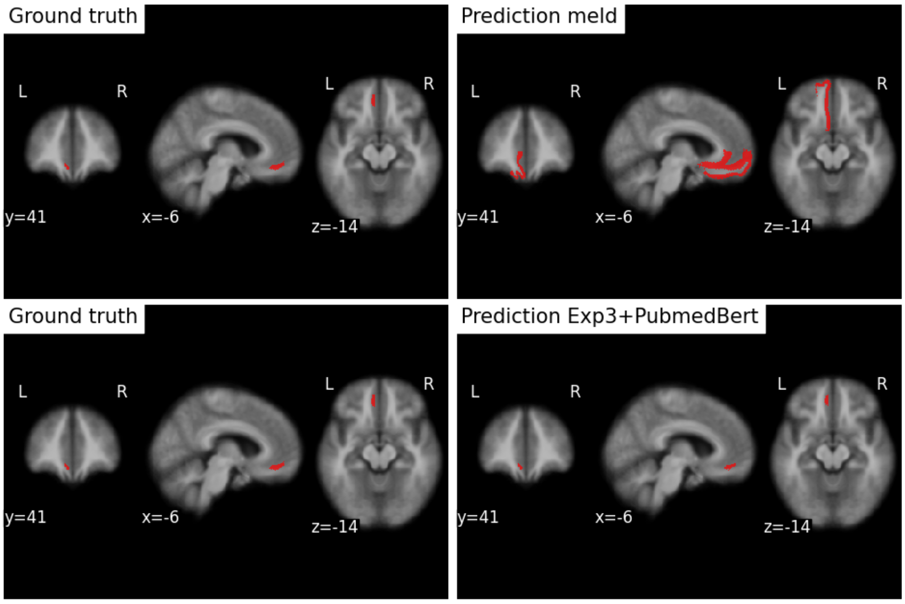


Figure 6.4: Comparison for subject `MELD_H3_3T_FCD_0002`.

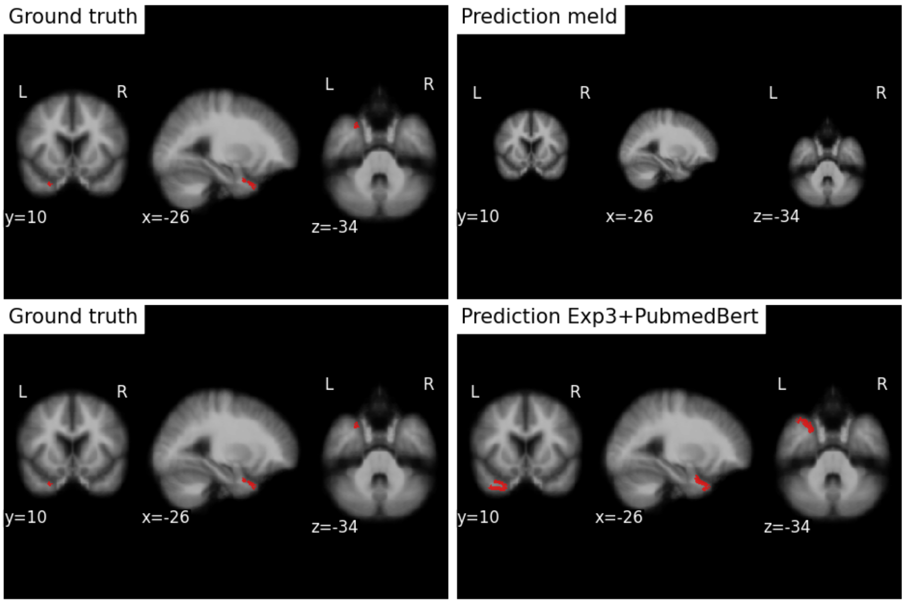


Figure 6.5: Comparison for subject `MELD_H16_3T_FCD_048`.

[1]Throughout this chapter, `Vision+LM` corresponds to Exp3.

Finally, we present an example comparing the same `Vision+LM` model conditioned on different textual descriptions, namely the *hemi* and *hemi + lobe* prompts.

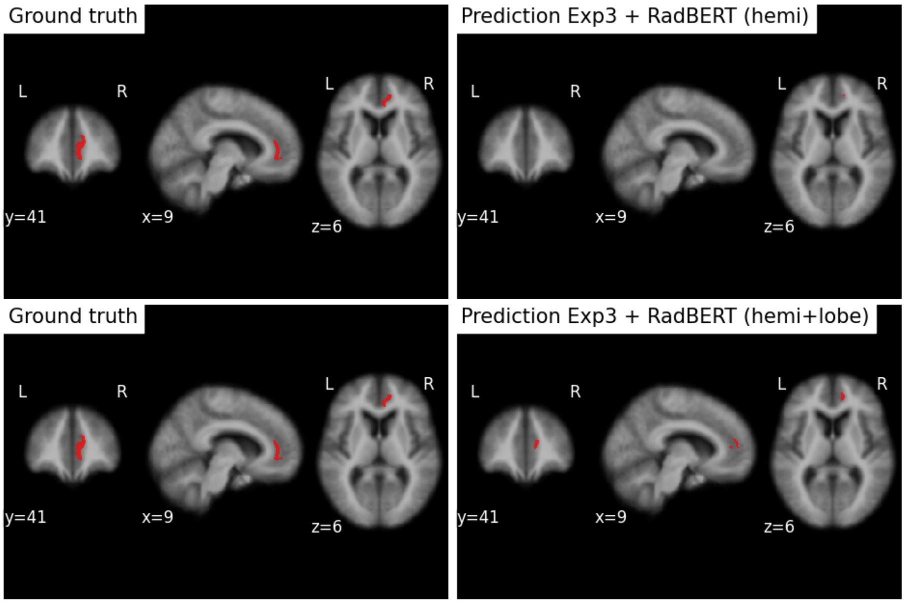


Figure 6.6: Comparison for subject `MELD_H3_3T_FCD_0018`.

6.2 Mixed Text

This chapter investigates training with *mixed text prompts* and the effect of a short decoder warm–up phase. We start from the `Vision+LM` architecture and compare two training schedules:

1. **Decoder open from the start**: the `GuideDecoder` is initialized with pretrained weights from *Vision–only* and optimized from the first epoch.

2. **Decoder warm–up (5 epochs)**: the model is initialized from the `Vision-only` checkpoint; during the first five epochs, we optimize only the text–fusion parameters in the `GuideDecoder` (self–/cross–attention and the text projection), while the MELD visual backbone, the text encoder (RadBERT), the SpiralConv–based upsampling path, and the segmentation head remain frozen. The choice of five epochs is pragmatic: each experiment uses an ensemble of models, training a single model takes roughly 8 hours, and typical runs span 20–30 epochs; therefore, a 5-epoch warm–up was considered sufficient.

We train the model with different atlas–based prompt variants, where one prompt type is randomly sampled per sample during training. All experiments in this chapter use **RadBERT** as the text encoder. Finally, we evaluate the trained models using intentionally mismatched (incorrect) prompts to assess robustness to prompt variability.

Warm–up vs. training from the start

Based on Tables 6.3 and 6.4, introducing a 5-epoch warm–up increases specificity by ∼8.8 p.p. and sensitivity by about 4.5 p.p. on average, while Dice, PPV, and IoU decrease slightly (by about 1–2 p.p.). We interpret this as evidence that the warm–up phase allows the `GuideDecoder` to first learn how to incorporate the textual prompts and establish stable cross–attention links to the visual embeddings, before updating the full model.

Table 6.3: Different text types for **Vision+LM_mixed** models (values in parentheses indicate 95% confidence intervals). *Decoder open from the start* — GuideDecoder trained from the first epoch. Main cohort — correct descriptions.

Model	Text type	Dice	PPV pixels	(mean) PPV clusters	IoU	Specificity, % (N = 193)	Sensitivity, % (N = 259)
MELD	–	0.230 (0.107–0.320)	0.166 (0.086–0.220)	0.515	0.130 (0.057–0.190)	58.0 (n=112) (51.3–64.8)	65.6 (n=170) (59.8–71.4)
Vision–only	–	0.238 (0.125–0.313)	0.198 (0.130–0.318)	0.361	0.135 (0.066–0.186)	31.1 (n=60) (24.9–37.8)	69.1 (n=179) (63.3–74.9)
Vision–only+SA	–	0.256 (0.125–0.326)	0.204 (0.122–0.322)	0.333	0.147 (0.067–0.195)	24.9 (n=48) (19.2–31.1)	70.7 (n=183) (65.3–76.1)
Vision+LM_mixed	hemi	0.235 (0.107–0.294)	0.222 (0.144–0.303)	0.511	0.133 (0.057–0.172)	79.3 (n=153) (73.6–85.0)	69.1 (n=179) (63.3–74.9)
Vision+LM_mixed	lobe_regions	0.243 (0.081–0.323)	0.246 (0.151–0.345)	0.491	0.138 (0.042–0.193)	79.3 (n=153) (73.6–85.0)	66.4 (n=172) (60.6–72.2)
Vision+LM_mixed	hemi+ lobe_regions	0.241 (0.097–0.312)	0.235 (0.152–0.346)	0.492	0.137 (0.051–0.185)	79.3 (n=153) (73.6–85.0)	68.7 (n=178) (62.9–74.1)
Vision+LM_mixed	hemi+lobe	0.241 (0.095–0.321)	0.233 (0.145–0.335)	0.505	0.137 (0.050–0.191)	79.3 (n=153) (73.6–85.0)	67.2 (n=174) (61.4–73.0)
Vision+LM_mixed	full_desc	0.251 (0.124–0.300)	0.256 (0.167–0.351)	0.479	0.144 (0.066–0.177)	79.3 (n=153) (73.6–85.0)	69.9 (n=181) (64.1–75.3)

Note: For healthy subjects (no lesion present), we use the same textual input for all prompt variants: *"No lesion detected."*

Table 6.4: Different text types for **Vision+LM_mixed freeze 5 epochs** models (values in parentheses indicate 95% confidence intervals). *Decoder warm-up for 5 epochs* — GuideDecoder unfrozen after epoch 5. Main cohort — correct descriptions.

Model	Text type	Dice	PPV pixels	(mean) PPV clusters	IoU	Specificity, % (N = 193)	Sensitivity, % (N = 259)
MELD	–	0.230 (0.107–0.320)	0.166 (0.086–0.220)	0.515	0.130 (0.057–0.190)	58.0 (n=112) (51.3–64.8)	65.6 (n=170) (59.8–71.4)
Vision–only	–	0.238 (0.125–0.313)	0.198 (0.130–0.318)	0.361	0.135 (0.066–0.186)	31.1 (n=60) (24.9–37.8)	69.1 (n=179) (63.3–74.9)
Vision–only+SA	–	0.256 (0.125–0.326)	0.204 (0.122–0.322)	0.333	0.147 (0.067–0.195)	24.9 (n=48) (19.2–31.1)	70.7 (n=183) (65.3–76.1)
Vision+LM_mixed	hemi	0.225 (0.131–0.294)	0.215 (0.137–0.325)	0.401	0.127 (0.070–0.172)	88.1 (n=170) (83.4–92.2)	73.7 (n=191) (68.3–79.2)
Vision+LM_mixed	lobe_regions	0.207 (0.123–0.330)	0.244 (0.160–0.307)	0.419	0.116 (0.065–0.197)	88.1 (n=170) (83.4–92.2)	71.8 (n=186) (66.4–77.2)
Vision+LM_mixed	hemi+ lobe_regions	0.219 (0.148–0.323)	0.216 (0.154–0.291)	0.392	0.123 (0.080–0.192)	88.1 (n=170) (83.4–92.2)	73.0 (n=189) (67.6–78.4)
Vision+LM_mixed	hemi+ lobe	0.228 (0.129–0.321)	0.247 (0.180–0.330)	0.482	0.129 (0.069–0.180)	88.1 (n=170) (83.4–92.2)	71.0 (n=184) (65.6–76.4)
Vision+LM_mixed	full_desc	0.230 (0.140–0.297)	0.204 (0.141–0.275)	0.342	0.130 (0.075–0.175)	88.1 (n=170) (83.4–92.2)	74.5 (n=193) (69.1–79.9)

Note: For healthy subjects (no lesion present), we use the same textual input for all prompt variants: *"No lesion detected."*

Wrong textual descriptions (Main cohort)

We now evaluate models with *incorrect* prompts, including the `no_text` setting (i.e., the model receives the placeholder string *full_brain* as input).

Across both Tables 6.5 and 6.6, models that were trained on correct textual descriptions but tested with intentionally incorrect ones still follow a consistent trend. Overall, when evaluated with intentionally incorrect prompts, performance decreases as expected, but the degradation is often modest and in some settings remains comparable to that obtained with correct descriptions. A possible explanation for this behavior is discussed in chapter 7.

Using the `no_text` setting (i.e., providing a fixed placeholder prompt, *full brain*, for both patients and healthy controls) does not substantially degrade overall performance; however, the metrics become consistently worse. Based on Table 6.5 and in comparison to evaluation with correct descriptions (Table 6.3), **Specificity** decreases by roughly 27 p.p.. A similar pattern is observed for **Sensitivity**: the `no_text` variant detects fewer lesions, with sensitivity lower by about 2 p.p. compared to using wrong prompts (Table 6.6), and by about 4 p.p. compared to using correct prompts (Table 6.3).

Without any linguistic prior, the attention becomes broad and less informative. Under strong class imbalance, the model behaves more conservatively and fires only at very high confidence.

Table 6.5: Different text types with *wrong* descriptions (without freezing). Main cohort — wrong descriptions

Model	**Text type**	**Dice**	**PPV pixels**	**(mean) PPV clusters**	**IoU**	**Specificity, %** (N = 193)	**Sensitivity, %** (N = 259)
MELD	–	0.230 (0.107–0.320)	0.166 (0.086–0.220)	0.515	0.130 (0.057–0.190)	58.0 (n=112) (51.3–64.8)	65.6 (n=170) (59.8–71.4)
Vision–only	–	0.238 (0.125–0.313)	0.198 (0.130–0.318)	0.361	0.135 (0.066–0.186)	31.1 (n=60) (24.9–37.8)	69.1 (n=179) (63.3–74.9)
Vision–only+SA	–	0.256 (0.125–0.326)	0.204 (0.122–0.322)	0.333	0.147 (0.067–0.195)	24.9 (n=48) (19.2–31.1)	70.7 (n=183) (65.3–76.1)
Vision+LM_mixed	wrong_hemi	0.234 (0.109–0.304)	0.217 (0.144–0.296)	0.509	0.133 (0.057–0.179)	79.3 (n=153) (73.6–85.0)	69.5 (n=180) (63.7–74.9)
Vision+LM_mixed	wrong_hemi + correct_lobe_regions	0.240 (0.097–0.312)	0.241 (0.152–0.346)	0.486	0.136 (0.051–0.185)	79.3 (n=153) (73.6–85.0)	67.6 (n=175) (61.8–73.4)
Vision+LM_mixed	wrong_hemi + correct_lobe	0.239 (0.088–0.317)	0.228 (0.146–0.332)	0.508	0.135 (0.046–0.188)	79.3 (n=153) (73.6–85.0)	67.6 (n=175) (61.8–73.4)
Vision+LM_mixed	wrong_hemi + wrong_lobe_regions	0.241 (0.098–0.320)	0.222 (0.147–0.343)	0.489	0.137 (0.052–0.190)	79.3 (n=153) (73.6–85.0)	68.0 (n=176) (62.2–73.7)
Vision+LM_mixed	wrong_hemi + wrong_lobe	0.243 (0.083–0.317)	0.234 (0.148–0.331)	0.498	0.139 (0.044–0.188)	79.3 (n=153) (73.6–85.0)	67.2 (n=174) (61.4–73.0)
Vision+LM_mixed	no_text	0.249 (0.080–0.296)	0.192 (0.127–0.289)	0.443	0.142 (0.042–0.174)	52.3 (n=101) (45.1–59.1)	66.0 (n=171) (60.2–71.8)

A similar effect is observed in Table 6.6. Even when models are *trained on correct textual descriptions* but evaluated using deliberately incorrect prompts, freezing the decoder for the first 5 epochs again leads to a performance gain. This effect is consistently observed across all settings with incorrect textual descriptions, except for the `no_text` baseline, where **Specificity** decreases by about 14 p.p. when compared to the corresponding no-freeze results in Table 6.5. In contrast, **Sensitivity** still improves by roughly 2 p.p. when decoder freezing is applied.

Table 6.6: Different text types with *wrong* descriptions (with 5 frozen epochs). Main cohort — wrong descriptions

Model	Text type	Dice	PPV pixels	(mean) PPV clusters	IoU	Specificity, % (N = 193)	Sensitivity, % (N = 259)
MELD	–	0.230 (0.107–0.320)	0.166 (0.086–0.220)	0.515	0.130 (0.057–0.190)	58.0 (n=112) (51.3–64.8)	65.6 (n=170) (59.8–71.4)
Vision–only	–	0.238 (0.125–0.313)	0.198 (0.130–0.318)	0.361	0.135 (0.066–0.186)	31.1 (n=60) (24.9–37.8)	69.1 (n=179) (63.3–74.9)
Vision–only+SA	–	0.256 (0.125–0.326)	0.204 (0.122–0.322)	0.333	0.147 (0.067–0.195)	24.9 (n=48) (19.2–31.1)	70.7 (n=183) (65.3–76.1)
Vision+LM_mixed	wrong_hemi	0.231 (0.119–0.287)	0.207 (0.139–0.303)	0.403	0.131 (0.063–0.167)	88.1 (n=170) (83.4–92.2)	73.7 (n=191) (68.3–79.2)
Vision+LM_mixed	wrong_hemi + correct_lobe_regions	0.216 (0.153–0.322)	0.216 (0.153–0.291)	0.395	0.121 (0.083–0.192)	88.1 (n=170) (83.4–92.2)	73.0 (n=189) (67.6–78.4)
Vision+LM_mixed	wrong_hemi + correct_lobe	0.222 (0.120–0.318)	0.243 (0.166–0.329)	0.479	0.125 (0.064–0.189)	88.1 (n=170) (83.4–92.2)	70.7 (n=183) (65.3–76.1)
Vision+LM_mixed	wrong_hemi+ wrong_lobe_regions	0.224 (0.135–0.330)	0.207 (0.149–0.298)	0.378	0.126 (0.072–0.198)	88.1 (n=170) (83.4–92.2)	74.1 (n=192) (68.7–79.5)
Vision+LM_mixed	wrong_hemi+ wrong_lobe	0.219 (0.114–0.331)	0.243 (0.164–0.346)	0.470	0.123 (0.060–0.198)	88.1 (n=170) (83.4–92.2)	71.0 (n=184) (65.6–76.4)
Vision+LM_mixed	no_text	0.245 (0.102–0.295)	0.205 (0.132–0.292)	0.384	0.140 (0.054–0.173)	38.3 (n=74) (31.6–45.1)	68.3 (n=177) (62.5–74.1)

Thus, the warm-up phase provides a consistent benefit: it prevents the decoder from immediately overfitting to noisy textual cues and encourages the model to extract whatever useful structural information is present, even when the text itself is incorrect.

Wrong textual descriptions (Independent cohort)

On the independent cohort (Tables 6.7–6.10), for both correct and incorrect prompts, the model with a 5-epoch decoder freeze generally detects ∼5 p.p. more lesions (higher **Sensitivity**). For most text–description settings, this increase in detected lesions does not substantially degrade segmentation quality: several segmentation metrics (**Dice**, **IoU**, and **PPV**) remain stable or even improve slightly compared to the non-frozen model. For the `hemi + lobe_regions` and `full_desc` prompts, we observe a noticeable decrease in segmentation quality (5–11 p.p. in Dice and 5–7 p.p. in IoU). However, these decreases are offset by a consistent improvement in lesion detection, which remains the primary objective in this setting.

Table 6.7: Different text types for **Vision+LM_mixed** models (values in parentheses indicate 95% confidence intervals). *Decoder open from the start* — GuideDecoder trained from the first epoch. Independent cohort — correct descriptions.

Model	Text type	Dice	PPV pixels	(mean) PPV clusters	IoU	Specificity, % (N = 83)	Sensitivity, % (N = 82)
MELD	–	0.358 (0.209–0.465)	0.261 (0.142–0.428)	0.464	0.218 (0.117–0.303)	55.4 (n=46) (44.6–66.3)	69.5 (n=57) (59.8–79.3)
Vision–only	–	0.376 (0.238–0.529)	0.443 (0.243–0.619)	0.350	0.232 (0.138–0.359)	39.8 (n=33) (28.9–50.6)	73.2 (n=60) (63.4–82.9)
Vision–only+SA	–	0.342 (0.155–0.494)	0.381 (0.177–0.651)	0.379	0.206 (0.084–0.328)	37.3 (n=31) (27.7–48.2)	69.5 (n=57) (59.8–79.3)
Vision+LM_mixed	hemi	0.430 (0.067–0.515)	0.429 (0.181–0.610)	0.604	0.274 (0.035–0.347)	89.2 (n=74) (81.9–95.2)	68.3 (n=56) (57.3–78.0)
Vision+LM_mixed	lobe_regions	0.374 (0.177–0.496)	0.486 (0.224–0.670)	0.656	0.230 (0.097–0.330)	89.2 (n=74) (81.9–95.2)	68.3 (n=56) (57.3–78.0)
Vision+LM_mixed	hemi+ lobe_regions	0.400 (0.222–0.501)	0.515 (0.242–0.668)	0.616	0.250 (0.125–0.334)	89.2 (n=74) (81.9–95.2)	69.5 (n=57) (59.8–79.3)
Vision+LM_mixed	hemi+lobe	0.380 (0.119–0.498)	0.515 (0.073–0.655)	0.613	0.234 (0.064–0.331)	89.2 (n=74) (81.9–95.2)	65.9 (n=54) (54.9–75.6)
Vision+LM_mixed	full_desc	0.402 (0.271–0.486)	0.515 (0.311–0.682)	0.626	0.251 (0.156–0.321)	89.2 (n=74) (81.9–95.2)	70.7 (n=58) (61.0–80.5)

Table 6.8: Different text types for **Vision+LM_mixed freeze 5 epochs** models (values in parentheses indicate 95% confidence intervals). *Decoder warm-up for 5 epochs* — GuideDecoder unfrozen after epoch 5. Independent cohort — correct descriptions.

Model	Text type	Dice	PPV pixels	(mean) PPV clusters	IoU	Specificity, % (N = 83)	Sensitivity, % (N = 82)
MELD	–	0.358 (0.209–0.465)	0.261 (0.142–0.428)	0.464	0.218 (0.117–0.303)	55.4 (n=46) (44.6–66.3)	69.5 (n=57) (59.8–79.3)
Vision–only	–	0.376 (0.238–0.529)	0.443 (0.243–0.619)	0.350	0.232 (0.138–0.359)	39.8 (n=33) (28.9–50.6)	73.2 (n=60) (63.4–82.9)
Vision–only+SA	–	0.342 (0.155–0.494)	0.381 (0.177–0.651)	0.379	0.206 (0.084–0.328)	37.3 (n=31) (27.7–48.2)	69.5 (n=57) (59.8–79.3)
Vision+LM_mixed	hemi	0.368 (0.248–0.505)	0.406 (0.218–0.596)	0.489	0.225 (0.141–0.338)	94.0 (n=78) (88.0–98.8)	74.4 (n=61) (64.6–84.1)
Vision+LM_mixed	lobe_regions	0.335 (0.144–0.474)	0.338 (0.168–0.543)	0.486	0.201 (0.077–0.311)	94.0 (n=78) (88.0–98.8)	72.0 (n=59) (62.2–81.7)
Vision+LM_mixed	hemi+ lobe_regions	0.332 (0.183–0.477)	0.343 (0.174–0.576)	0.474	0.199 (0.101–0.313)	94.0 (n=78) (88.0–98.8)	75.6 (n=62) (65.9–84.1)
Vision+LM_mixed	hemi+lobe	0.400 (0.162–0.507)	0.462 (0.239–0.672)	0.586	0.250 (0.088–0.340)	94.0 (n=78) (88.0–98.8)	74.4 (n=61) (64.6–84.1)
Vision+LM_mixed	full_desc	0.305 (0.210–0.478)	0.326 (0.164–0.651)	0.306	0.180 (0.117–0.314)	94.0 (n=78) (88.0–98.8)	78.0 (n=64) (68.3–86.6)

Table 6.9: Different text types with *wrong* descriptions (without freezing). Independent cohort — wrong descriptions

Model	Text type	Dice	PPV pixels	(mean) PPV clusters	IoU	Specificity, % (N = 83)	Sensitivity, % (N = 82)
MELD	–	0.358 (0.209–0.465)	0.261 (0.142–0.428)	0.464	0.218 (0.117–0.303)	55.4 (n=46) (44.6–66.3)	69.5 (n=57) (59.8–79.3)
Vision–only	–	0.376 (0.238–0.529)	0.443 (0.243–0.619)	0.350	0.232 (0.138–0.359)	39.8 (n=33) (28.9–50.6)	73.2 (n=60) (63.4–82.9)
Vision–only+SA	–	0.342 (0.155–0.494)	0.381 (0.177–0.651)	0.379	0.206 (0.084–0.328)	37.3 (n=31) (27.7–48.2)	69.5 (n=57) (59.8–79.3)
Vision+LM_mixed	wrong_hemi	0.421 (0.067–0.520)	0.432 (0.179–0.609)	0.604	0.267 (0.035–0.352)	89.2 (n=74) (81.9–95.2)	68.3 (n=56) (57.3–78.0)
Vision+LM_mixed	wrong_hemi + correct_lobe_regions	0.392 (0.214–0.503)	0.515 (0.240–0.675)	0.625	0.244 (0.120–0.336)	89.2 (n=74) (81.9–95.2)	69.5 (n=57) (59.8–79.3)
Vision+LM_mixed	wrong_hemi + correct_lobe	0.378 (0.119–0.492)	0.509 (0.073–0.658)	0.613	0.233 (0.063–0.326)	89.2 (n=74) (81.9–95.2)	65.9 (n=54) (54.9–75.6)
Vision+LM_mixed	wrong_hemi + wrong_lobe_regions	0.433 (0.258–0.515)	0.475 (0.256–0.676)	0.660	0.276 (0.148–0.346)	89.2 (n=74) (81.9–95.2)	69.5 (n=57) (59.8–79.3)
Vision+LM_mixed	wrong_hemi + wrong_lobe	0.388 (0.147–0.501)	0.481 (0.154–0.672)	0.617	0.241 (0.079–0.334)	89.2 (n=74) (81.9–95.2)	67.1 (n=55) (56.1–76.8)
Vision+LM_mixed	no_text	0.397 (0.138–0.502)	0.507 (0.151–0.647)	0.549	0.248 (0.074–0.335)	68.7 (n=57) (59.0–78.3)	65.9 (n=54) (54.9–75.6)

We also compare two key text–description types, `hemi+lobe` and `hemi+lobe_regions`, under both training strategies (with and without decoder warm-up). For models trained *without* decoder warm-up, the `hemi+lobe_regions` setting consistently produces higher segmentation quality: Dice and IoU improve by ∼2-3 p.p., and Sensitivity increases by ∼2-4 p.p. compared to `hemi+lobe`. The same trend is seen in the evaluation of the incorrect text condition, where we compare `wrong_hemi + wrong_lobe` with `wrong_hemi + wrong_lobe_regions`. However, for models trained *with* 5-epoch warm-up, the situation is the opposite. In this case, the `hemi+lobe` model provides higher segmentation accuracy, outperforming `hemi+lobe_regions` by ∼3-7 p.p. in Dice and IoU, although with a slight decrease in Sensitivity (typically 1-2 p.p.).

Table 6.10: Different text types with *wrong* descriptions (with 5 frozen epochs). Independent cohort — wrong descriptions

Model	Text type	Dice	PPV pixels	(mean) PPV clusters	IoU	Specificity, % (N = 83)	Sensitivity, % (N = 82)
MELD	–	0.358 (0.209–0.465)	0.261 (0.142–0.428)	0.464	0.218 (0.117–0.303)	55.4 (n=46) (44.6–66.3)	69.5 (n=57) (59.8–79.3)
Vision–only	–	0.376 (0.238–0.529)	0.443 (0.243–0.619)	0.350	0.232 (0.138–0.359)	39.8 (n=33) (28.9–50.6)	73.2 (n=60) (63.4–82.9)
Vision–only+SA	–	0.342 (0.155–0.494)	0.381 (0.177–0.651)	0.379	0.206 (0.084–0.328)	37.3 (n=31) (27.7–48.2)	69.5 (n=57) (59.8–79.3)
Vision+LM_mixed	wrong_hemi	0.368 (0.273–0.499)	0.382 (0.220–0.558)	0.471	0.225 (0.158–0.333)	94.0 (n=78) (88.0–98.8)	74.4 (n=61) (64.6–84.1)
Vision+LM_mixed	wrong_hemi + correct_lobe_regions	0.321 (0.163–0.456)	0.336 (0.152–0.545)	0.471	0.192 (0.089–0.296)	94.0 (n=78) (88.0–98.8)	74.4 (n=61) (64.6–84.1)
Vision+LM_mixed	wrong_hemi + correct_lobe	0.401 (0.129–0.507)	0.448 (0.240–0.676)	0.578	0.253 (0.069–0.340)	94.0 (n=78) (88.0–98.8)	74.4 (n=61) (64.6–84.1)
Vision+LM_mixed	wrong_hemi + wrong_lobe_regions	0.325 (0.196–0.458)	0.338 (0.205–0.619)	0.444	0.194 (0.109–0.297)	94.0 (n=78) (88.0–98.8)	74.4 (n=61) (64.6–84.1)
Vision+LM_mixed	wrong_hemi + wrong_lobe	0.353 (0.137–0.489)	0.374 (0.245–0.622)	0.551	0.214 (0.074–0.324)	94.0 (n=78) (88.0–98.8)	72.0 (n=59) (62.2–81.7)
Vision+LM_mixed	no_text	0.345 (0.108–0.453)	0.349 (0.156–0.567)	0.406	0.209 (0.058–0.292)	47.0 (n=39) (36.1–57.8)	64.6 (n=53) (53.7–74.4)

These results suggest that the warm-up strategy is preferable for practical use. In real clinical workflows, coarse anatomical labels are far more common than precise region-level descriptions, and in such scenarios the warm-up model provides more reliable segmentation accuracy, even if this comes with a slight reduction in the number of detected lesions.

6.3 Linking MELD to GNN

In these experiments, we investigated the effect of connecting different numbers of MELD feature stages to the *GNN blocks*.

On the main cohort (Table 6.11), the configuration with **three GNN layers** provides the most balanced performance across Dice, PPV_{pixels}, and IoU, while also detecting **two additional lesions** compared to the no-GNN baseline. This is the only configuration that consistently ranks among the top performers across all primary metrics, supporting its selection as the best overall trade-off between lesion detection and false positive control.

Connecting **all seven** MELD stages to the GNN yields the highest lesion count, with sensitivity increasing by ∼3.5 p.p. (188/259 vs. 179/259). This improvement comes with only a modest reduction in Dice and IoU (on the order of 2–3 p.p.). Such behavior is consistent with the properties of SAGEConv-based neighborhood aggregation: increasing GNN depth expands each node's receptive field, while incorporating multi-stage MELD features injects complementary local and global contextual information. Together, these effects enable the detection of additional lesion clusters that may be missed when relying solely on local features.

Specificity shows a non-uniform behavior across configurations. Connecting GNN blocks to the first one or two MELD feature stages results in a substantial increase in specificity (approximately +18 and +9 p.p., respectively) compared to the no-GNN baseline. In contrast, further increasing the number of GNN layers leads to a gradual decrease in specificity. One possible explanation is that deeper message passing propagates lesion-related activations beyond the actual lesion regions, increasing the likelihood of assigning lesion–like predictions to anatomically normal cortical regions in healthy controls. This effect is particularly evident for later MELD stages, where features are lower-dimensional and the graph becomes denser, making it harder to preserve spatial distinctions. Consequently, the model favors higher sensitivity at the expense of specificity, resulting in an increased false-positive rate.

Table 6.11: Different number of MELD stages connected to the GNN block (values in parentheses indicate 95% confidence intervals). Main cohort

Experiment	Dice	PPV pixels	(mean) PPV clusters	IoU	Specificity, % (N = 193)	Sensitivity, % (N = 259)
Vision–only: 0 layers	0.238 (0.125–0.313)	0.198 (0.130–0.318)	0.361	0.135 (0.066–0.186)	31.1 (n=60) (24.9–37.8)	69.1 (n=179) (63.3–74.9)
Vision–only: 1 layer	0.234 (0.093–0.313)	0.236 (0.149–0.340)	0.422	0.133 (0.049–0.186)	49.2 (n=95) (42.5–56.5)	67.2 (n=174) (61.4–73.0)
Vision–only: 2 layers	0.234 (0.126–0.334)	0.219 (0.136–0.371)	0.395	0.132 (0.067–0.200)	40.9 (n=79) (34.2–48.2)	68.3 (n=177) (62.5–74.1)
Vision–only: 3 layers	0.249 (0.114–0.332)	0.224 (0.142–0.341)	0.285	0.142 (0.060–0.199)	16.1 (n=31) (10.9–21.2)	69.9 (n=181) (64.1–75.3)
Vision–only: 4 layers	0.231 (0.097–0.304)	0.180 (0.128–0.283)	0.287	0.130 (0.051–0.179)	2.1 (n=4) (0.5–4.1)	68.3 (n=177) (62.5–74.1)
Vision–only: 5 layers	0.221 (0.111–0.314)	0.220 (0.153–0.326)	0.268	0.123 (0.059–0.187)	2.6 (n=5) (0.5–5.2)	71.4 (n=185) (66.0–76.8)
Vision–only: 6 layers	0.208 (0.121–0.312)	0.219 (0.147–0.336)	0.312	0.116 (0.065–0.185)	11.9 (n=23) (7.8–16.6)	69.1 (n=179) (63.3–74.9)
Vision–only: 7 layers	0.206 (0.119–0.293)	0.219 (0.135–0.341)	0.210	0.115 (0.063–0.172)	1.0 (n=2) (0.0–2.6)	72.6 (n=188) (67.2–78.0)

On the independent (Bonn) cohort (Table 6.12), the **5-layer** configuration provides the best overall trade-off. It achieves the highest Dice and IoU scores while also detecting the largest number of lesions (64/82). Compared to the no-GNN baseline (0 layers), this corresponds to an absolute improvement of ∼10.4 p.p. in Dice and ∼8.3 p.p. in IoU, highlighting the benefit of incorporating the proposed GNN block. At the same time, this configuration exhibits very low specificity, consistent with the trends observed on the main cohort and discussed earlier.

For subsequent experiments we keep the **3-layer** and **5-layer** variants, and will compare them in the final study to determine the optimal number of connected MELD stages.

Table 6.12: Different number of MELD stages connected to the GNN block (values in parentheses indicate 95% confidence intervals). Independent cohort (Bonn Dataset)

Experiment	Dice	PPV pixels	(mean) PPV clusters	IoU	Specificity, % (N = 83)	Sensitivity, % (N = 82)
Vision–only: 0 layers	0.376 (0.238–0.529)	0.443 (0.243–0.619)	0.350	0.232 (0.138–0.359)	39.8 (n=33) (28.9–50.6)	73.2 (n=60) (63.4–82.9)
Vision–only: 1 layer	0.339 (0.189–0.507)	0.590 (0.278–0.750)	0.496	0.204 (0.104–0.340)	59.0 (n=49) (48.2–69.9)	69.5 (n=57) (59.8–79.3)
Vision–only: 2 layers	0.383 (0.216–0.506)	0.432 (0.243–0.663)	0.477	0.237 (0.123–0.339)	55.4 (n=46) (44.6–66.3)	70.7 (n=58) (61.0–80.5)
Vision–only: 3 layers	0.390 (0.302–0.517)	0.503 (0.309–0.732)	0.311	0.242 (0.178–0.349)	16.9 (n=14) (9.6–25.3)	75.6 (n=62) (65.9–84.1)
Vision–only: 4 layers	0.458 (0.279–0.600)	0.491 (0.186–0.582)	0.298	0.297 (0.162–0.428)	9.6 (n=8) (3.6–15.7)	73.2 (n=60) (63.4–82.9)
Vision–only: 5 layers	0.480 (0.339–0.589)	0.483 (0.289–0.581)	0.240	0.315 (0.204–0.417)	4.8 (n=4) (1.2–9.6)	78.0 (n=64) (68.3–86.6)
Vision–only: 6 layers	0.382 (0.292–0.544)	0.584 (0.342–0.687)	0.344	0.236 (0.171–0.373)	30.1 (n=25) (20.5–39.8)	73.2 (n=60) (63.4–82.9)
Vision–only: 7 layers	0.402 (0.284–0.493)	0.375 (0.232–0.634)	0.206	0.252 (0.165–0.327)	1.2 (n=1) (0.0–3.6)	74.4 (n=61) (64.6–84.1)

6.4 Final Experiments

Based on the observations from the previous chapters, we proceed with a set of follow–up experiments using the full multimodal architecture with GNN blocks. Specifically, we consider the following experimental setup:

(i) the text encoder is RadBERT, which is used consistently throughout all experiments in this work;

(ii) the visual branch employs GNN encoders with 3 and 5 blocks, corresponding to the settings analyzed in the vision–only experiments; and

(iii) for each GNN depth, the model is initialized from the corresponding `Vision-only`+GNN checkpoint and trained using a 5–epoch decoder warm–up, after which the full decoder is unfrozen.

For these experiments, we will use the following types of text prompts: "hemisphere" and "hemisphere + lobe". Using "lobe regions" and "full text" is not meaningful, as discussed earlier. We have also included a "no text" column, where instead of an empty string, this column will contain a general description: "full brain". Additionally, we will train the model with mixed text, primarily to reduce the overall experimental runtime, since our goal here is to demonstrate relative improvements under optimal settings, rather than to obtain the best possible model.

The architecture used in this experiment for `Vision-only` is shown below:

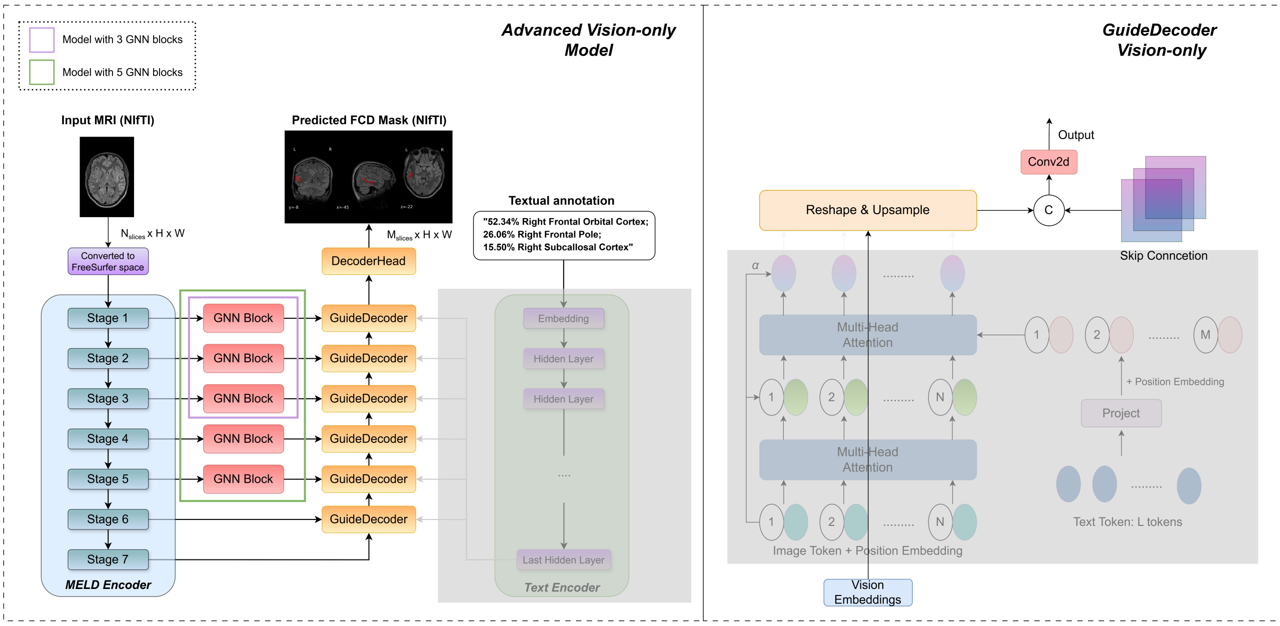


Figure 6.7: Architecture of Advanced `Vision-only`: MELD surface-based features are extracted through a multi-stage GNN encoder (3- or 5-block variants) and passed through a hierarchical GuideDecoder with cross-attention to text embeddings. At each decoder stage, visual features are fused with text embeddings, reshaped, and upsampled through geometry-aware operations, before producing the final segmentation via the DecoderHead. The right panel illustrates the internal structure of the GuideDecoder layer, including self- and cross-attention over text and vision tokens, skip connections, and spatial upsampling.

On the main cohort (Table 6.13), in the *hemisphere + lobe* and *no_text* settings, the models with GNN blocks indeed detect a larger number of lesions. However, in most cases the model without a GNN block exhibits higher values across all primary metrics and clearly dominates in terms of *Specificity*.

The $\text{PPV}_{\text{clusters}}$ metric explicitly indicates that GNN-based models produce substantially more false-positive clusters, which negatively affects the overall prediction quality. Therefore, based on the obtained results, no definitive advantage of information aggregation in sparse graphs can be concluded.

Table 6.13: Median performance on the main cohort (with 95% confidence intervals).

Model	Text type	Deocder type	Dice	PPV pixels	(mean) PPV clusters	IoU	Specificity, % (N = 193)	Sensitivity, % (N = 259)
MELD	–	–	0.230 (0.107–0.320)	0.166 (0.086–0.220)	0.515	0.130 (0.057–0.190)	58.0 (n=112) (51.3–64.8)	65.6 (n=170) (59.8–71.4)
Vision+LM_mixed	hemisphere	Vision–only with 3 GNN	**0.232** (0.141-0.312)	0.212 (0.146-0.330)	0.314	**0.131** (0.076-0.185)	8.3 (n=16) (4.7–12.4)	69.9 (n=181) (64.1–75.3)
		Vision–only with 5 GNN	0.211 (0.110–0.320)	**0.226** (0.154–0.394)	0.242	0.118 (0.058–0.191)	1.0 (n=2) (0.0–2.6)	72.2 (n=187) (66.8–77.6)
		Basic Vision–only	0.225 (0.131–0.294)	0.215 (0.137–0.325)	**0.401**	0.127 (0.070–0.172)	**88.1 (n=170)** (83.4–92.2)	**73.7 (n=191)** (68.3–79.2)
Vision+LM_mixed	hemisphere + lobe	Vision–only with 3 GNN	0.224 (0.154-0.319)	0.200 (0.142-0.314)	0.310	0.126 (0.083-0.190)	8.3 (n=16) (4.7–12.4)	69.5 (n=180) (63.7–74.9)
		Vision–only with 5 GNN	0.186 (0.119–0.314)	0.241 (0.158–0.384)	0.246	0.103 (0.063–0.186)	1.0 (n=2) (0.0–2.6)	**71.8 (n=186)** (66.4–77.2)
		Basic Vision–only	**0.228** (0.129–0.321)	**0.247** (0.180–0.330)	**0.482**	**0.129** (0.069–0.180)	**88.1 (n=170)** (83.4–92.2)	71.0 (n=184) (65.6–76.4)
Vision+LM_mixed	no text	Vision–only with 3 GNN	0.212 (0.120-0.311)	**0.262** (0.164-0.355)	0.206	0.119 (0.064-0.184)	0.0 (n=0) (0.0–0.0)	**72.2 (n=187)** (66.8–77.6)
		Vision–only with 5 GNN	0.229 (0.118–0.311)	0.259 (0.135–0.363)	0.235	0.130 (0.063–0.184)	1.0 (n=2) (0.0–2.6)	69.5 (n=180) (63.7–74.9)
		Basic Vision–only	**0.249** (0.080–0.296)	0.192 (0.127–0.289)	**0.443**	**0.142** (0.042–0.174)	**52.3 (n=101)** (45.1–59.1)	66.0 (n=171) (60.2–71.8)

Note.
Basic Vision–only denotes the pretrained `Vision-only` model without GNN blocks; *Vision–only with 3/5 GNN* denotes the pretrained `Vision-only` model with 3/5 GNN blocks.
Dice, IoU, and PPV metrics are computed on patients only (subjects with lesions).
All experiments in this table use the *Vision+LM_mixed* training scheme to obtain results more efficiently.

However, on the independent (Bonn) cohort (Tables 6.14), we observe that for the *hemisphere* and *hemisphere + lobe* settings, the model with **three GNN blocks** achieves the highest Dice, IoU, and Sensitivity. However, its $\text{PPV}_{\text{clusters}}$ is almost twice as low as that of the baseline without a GNN block, indicating a substantially larger number of false-positive clusters. This is further reflected in the **Specificity**, which is approximately three times lower than the baseline and substantially lower compared to the 5-GNN configuration.

These findings suggest that the number of GNN blocks must be selected with caution: deeper aggregation can cause feature oversmoothing, which may negatively affect cluster-level precision, as clearly seen in the 5-GNN case. In the *no-text* setting, GNN-based models primarily increase the number of detected lesions, but do so at the cost of all remaining metrics, indicating reduced prediction quality.

Table 6.14: Median performance on the independent cohort (with 95% confidence intervals).

Model	Text type	Deocder type	Dice	PPV pixels	(mean) PPV clusters	IoU	Specificity, % (N = 83)	Sensitivity, % (N = 82)
MELD	–	–	0.358 (0.209–0.465)	0.261 (0.142–0.428)	0.464	0.218 (0.117–0.303)	55.4 (n=46) (44.6–66.3)	69.5 (n=57) (59.8–79.3)
Vision+LM_mixed	hemisphere	Vision–only with 3 GNN	**0.446** (0.289–0.553)	0.404 (0.228–0.605)	0.294	**0.288** (0.169–0.382)	33.7 (n=28) (24.1–44.6)	**78.0 (n=64)** (68.3–86.6)
		Vision–only with 5 GNN	0.378 (0.274–0.549)	**0.418** (0.237–0.667)	0.186	0.233 (0.159–0.379)	1.2 (n=1) (0.0–3.6)	75.6 (n=62) (65.9–84.1)
		Basic Vision–only	0.368 (0.248–0.505)	0.406 (0.218–0.596)	**0.489**	0.225 (0.141–0.338)	**94.0 (n=78)** (88.0–98.8)	74.4 (n=61) (64.6–84.1)
Vision+LM_mixed	hemisphere + lobe	Vision–only with 3 GNN	**0.412** (0.302–0.565)	0.388 (0.217–0.567)	0.299	**0.260** (0.178–0.394)	33.7 (n=28) (24.1–44.6)	**78.0 (n=64)** (68.3–86.6)
		Vision–only with 5 GNN	0.379 (0.259-0.541)	0.408 (0.220-0.666)	0.188	0.233 (0.148-0.371)	1.2 (n=1) (0.0–3.6)	75.6 (n=62) (65.9–84.1)
		Basic Vision–only	0.400 (0.162–0.507)	**0.462** (0.239–0.672)	**0.586**	0.250 (0.088–0.340)	**94.0 (n=78)** (88.0–98.8)	74.4 (n=61) (64.6–84.1)
Vision+LM_mixed	no text	Vision–only with 3 GNN	0.299 (0.161-0.431)	0.385 (0.193-0.711)	0.202	0.176 (0.088-0.274)	1.2 (n=1) (0.0–3.6)	72.0 (n=59) (62.2–81.7)
		Vision–only with 5 GNN	0.342 (0.242-0.464)	0.376 (0.215-0.676)	0.191	0.206 (0.138-0.302)	0.0 (n=0) (0.0–0.0)	**74.4 (n=61)** (64.6–84.1)
		Basic Vision–only	**0.397** (0.138–0.502)	**0.507** (0.151–0.647)	**0.549**	**0.248** (0.074–0.335)	**68.7 (n=57)** (59.0–78.3)	65.9 (n=54) (54.9–75.6)

Note.
Basic Vision–only denotes the pretrained `Vision-only` model without GNN blocks; *Vision–only with 3/5 GNN* denotes the pretrained `Vision-only` model with 3/5 GNN blocks.
Dice, IoU, and PPV metrics are computed on patients only (subjects with lesions).
All experiments in this table use the *Vision+LM_mixed* training scheme to obtain results more efficiently.

Overall, the results show that additional message passing can indeed increase sensitivity for sparse graphs by capturing more lesion candidates, but this benefit comes with a pronounced increase in false positives. Therefore, while deeper aggregation offers potential advantages, it also highlights the need for further refinement of GNN-based feature integration to avoid sacrificing precision.

7 Discussion

The experiments conducted in this thesis demonstrate that incorporating textual information into surface-based GNN architectures can substantially improve the detection of FCD type II. At the same time, the results highlight important trade-offs that must be considered when designing multimodal models, particularly with respect to the balance between sensitivity and specificity.

Impact of textual guidance in the basic experiments

The basic experiments show that the form and granularity of textual input play a crucial role in multimodal lesion detection. Among all configurations, **Vision+LM_hemi+lobe_regions** achieves the most balanced performance while providing the highest sensitivity. Incorporating hemisphere- and lobe-region information constrains the search space, leading to improved lesion localization.

The **Vision+LM_hemi+lobe+PubMedBERT** configuration yields the highest overall sensitivity and specificity across both cohorts. The concise prompt without irrelevant context, combined with PubMedBERT's domain-specific pretraining, improves detection performance while only slightly reducing segmentation metrics such as Dice, IoU, PPV.

In contrast, the **Vision+LM_full_desc** variant underperforms relative to the hemi+lobe-based prompts. This suggests that long atlas-style descriptions introduce redundancy and noise, whereas shorter, more targeted prompts provide a clearer guidance signal to the model.

Role of GNN-based aggregation

Ahe analysis of experiments linking MELD features to a GNN block reveals a nuanced picture. Aggregating information from neighboring nodes improves the model's ability to detect lesions, as reflected by increased sensitivity and lesion-level coverage. This indicates that message passing can amplify subtle abnormal patterns that may be missed by purely local representations.

However, this benefit comes at the cost of reduced specificity, manifested as an increased false-positive rate on healthy controls. This is an important limitation for clinical use, since our evaluation does not include other epilepsy-related abnormalities or non-FCD lesion types. Therefore, we cannot assess how often the model would spuriously highlight regions in patients with different pathologies.

Importantly, increasing the depth of GNN propagation does not lead to monotonic performance improvements. For higher-dimensional and sparser MELD feature stages, neighborhood aggregation stabilizes representations and enriches contextual information. In contrast, for lower-dimensional and denser stages (e.g., stages 6–7), additional message passing induces oversmoothing, blurring spatial distinctions and degrading performance.

Influence of mixed textual input and warm-up training

Experiments with mixed textual descriptions demonstrate that even coarse prompts, such as hemisphere- or lobe-only descriptions, can meaningfully improve lesion detection compared to purely vision-based baselines. Across configurations, gains in sensitivity typically come with a trade-off in specificity, i.e., an increased number of false positive clusters. Overall, reduced–text prompts (e.g., hemisphere + lobe) often provide a more favorable balance between detecting lesions and limiting false positives, which may be preferable for clinical use.

The warm-up experiments further indicate that freezing the visual backbone during the initial training epochs can be beneficial. A plausible explanation is that this strategy reduces sensitivity to prompt variability and mildly incorrect descriptions. This behavior can be attributed to three factors: (i) even imperfect prompts convey coarse structural information that narrows the search space, (ii) the decoder exhibits robustness to mild text–image mismatch due to pretraining, and (iii) training with slightly noisy prompts acts as a regularizer, discouraging overfitting to specific wording and promoting more conservative predictions.

Observations from appendix analyses

The text distribution analysis shows that conditioning on fine-grained region names tends to induce overfitting due to the highly imbalanced and long-tailed label distribution. Frequent regions dominate the learning signal, while rare categories lack sufficient variability for robust generalization.

The semantic similarity analysis further highlights the importance of the text encoder. Biomedical language models pretrained on domain-specific corpora (e.g., PubMedBERT, RadBERT, BlueBERT) exhibit higher semantic coherence between anatomically related brain regions. This suggests that semantically aligned text embeddings facilitate more effective multimodal fusion and improve both interpretability and performance.

General Observations

Although the observed improvements in sensitivity and lesion-level detection are consistent across experiments, the results should be interpreted as a methodological proof of concept demonstrating the benefit of fusing visual and textual information, rather than as a clinically deployable solution. In this work, textual input is limited to structured, atlas-derived descriptions with explicit anatomical labels. Importantly, the performance gains cannot be attributed to the text providing explicit supervision about lesion presence: for healthy subjects, the text indicates “no lesion detected”, whereas for patients with FCD, including the “wrong text” setups, the text only provides coarse spatial information (e.g., hemisphere or lobe) and does not encode whether a lesion is present. The main effect of textual guidance is therefore a spatial regularization of the search space, which can improve sensitivity but may also increase false positives when the anatomical prior is inaccurate.

Limitations and future directions

Several limitations must be acknowledged. While the MELD surface features are ComBat-harmonized to reduce inter-site effects, residual variability may remain due to differences in acquisition protocols, preprocessing, and lesion annotation practices across centers. Furthermore, the MELD cohort predominantly consists of histologically confirmed type II FCD cases, and it therefore remains unclear whether the same conclusions extend to other FCD subtypes. Finally, the model relies on atlas-based text generated automatically; incorporating free-form radiology reports could yield a more realistic estimate of performance under clinical conditions, where textual descriptions are not always precise.

This work does not explore the effect of unfreezing different numbers of the language model layers or varying the number of cross–modal connections to the GuideDecoder. However, previous work (Huemann et al., 2024 [31]) shows that selectively unfreezing specific layers of a language model and fine-tuning them for a segmentation task can yield consistent performance gains. These aspects remain for future investigation.

Another conceptual direction concerns the representation of anatomical descriptions. In this thesis, atlas information was encoded via textual prompts and processed by a pretrained language model. An alternative would be to express the same information as a numerical feature vector with one dimension per cortical region. In such a vector, the percentages mentioned in the textual description (e.g., 52.34% Right Frontal Orbital Cortex; 26.06% Right Frontal Pole) would directly populate the corresponding entries, while all other regions would be set to zero. For coarse labels such as lobe-only or hemisphere-only descriptions, the corresponding weight could be uniformly distributed across all constituent regions.

While this vector–based formulation is simple, interpretable, and avoids the computational overhead of a pretrained LM, it is also less expressive in its basic form. When used without an explicitly defined structure, a model trained solely on such numerical representations treats each vector dimension as an independent feature and does not explicitly encode anatomical relationships, such as lobar or hemispheric grouping, spatial adjacency, or hierarchical organization. Although these relationships could in principle be incorporated through carefully designed numerical encodings or structured representations, this requires additional modeling assumptions and design choices. In contrast, language models implicitly encode semantic relationships between anatomical terms, including regional similarity and hierarchical structure, and naturally handle coarse, uncertain, or partially specified descriptions. This makes LM–based representations particularly suitable for prototyping multimodal fusion and exploring how textual guidance can complement visual features. Accordingly, while numerical vector representations remain a sustainable baseline, their relative performance compared to LM-based approaches depends on how explicitly anatomical structure and uncertainty are modeled, and remains an open question for future work.

Overall, the results indicate that textual guidance is a promising direction for FCD detection. However, the balance between sensitivity and precision remains a critical design choice that must be adapted to the clinical context and intended use case.

8 Conclusion

This thesis introduced a novel multimodal segmentation framework for detecting focal cortical dysplasia type II, combining surface-based visual representations with textual priors derived from anatomical atlases. The main findings are summarized below:

- Incorporating textual features consistently improved segmentation performance over vision-only baselines, demonstrating the value of multimodal fusion in low-data medical imaging scenarios.
- Aggregating MELD features using a GNN block increased the number of detected lesions but also introduced a higher rate of false positives. While this indicates that additional message passing can enhance sensitivity, it also shows that the current aggregation strategy requires further refinement to avoid degrading overall quality.
- Even coarse textual labels such as hemisphere- or lobe-level descriptions yielded substantial improvements. This suggests that fine-grained region names are not strictly necessary, and that lightweight, structured prompts may offer a more robust and generalizable alternative to detailed atlas-style descriptions.

Overall, the results demonstrate that multimodal architectures integrating language and surface-based vision are both feasible and beneficial for epileptogenic lesion detection under limited-data conditions. This work represents an initial step toward leveraging anatomical language priors in surface-based segmentation models.

Future research should explore training on larger and more heterogeneous cohorts, incorporating clinical radiology reports as richer textual supervision, and investigating alternative fusion strategies or graph-based aggregation mechanisms to improve robustness, reduce false positives, and enhance generalizability across clinical centers.

Appendix

HexUnpool

Surface upsampling is performed using the `HexUnpool` operator introduced in the MELD framework [9]. Given features $X \in \mathbb{R}^{B \times H \times N_{\text{from}} \times C}$ defined on a coarse icosphere with N_{from} vertices, the operator produces an upsampled tensor $X' \in \mathbb{R}^{B \times H \times N_{\text{to}} \times C}$ on a denser icosphere ($N_{\text{to}} > N_{\text{from}}$).

The icosphere hierarchy used in MELD is generated in a deterministic manner: each refinement step subdivides the triangular mesh, creating new vertices at midpoints of edges while preserving the spherical topology. For every fine-level vertex, the MELD preprocessing pipeline provides a precomputed list of its *parent* vertices on the coarser level. These mappings are stored in the accompanying icosphere files (e.g., `coords`, `neighbours`, `spirals`, `t_edges`) and define how features should be propagated across resolutions. We directly use these mappings to ensure geometrically consistent upsampling.

During upsampling, two types of vertices must be handled:

1. **Vertices inherited from the coarse mesh.** Their features are copied directly to the corresponding positions in X'.

2. **Newly introduced vertices.** Each fine-level vertex v corresponds to one or more coarse-level *parent* vertices. Let $\mathcal{I}(v)$ denote the (predefined) set of indices of parent vertices from which v was generated during mesh refinement. These are not geometric neighbors on the fine mesh but a fixed upsampling correspondence determined by the icosphere construction. The feature of v is computed as the mean of its parents:

$$X'_v = \frac{1}{|\mathcal{I}(v)|} \sum_{u \in \mathcal{I}(v)} X_u.$$

This yields a smooth and topology-consistent interpolation of features from the coarse surface to the finer one. Since the mapping $\mathcal{I}(v)$ is deterministically defined by the MELD icosphere hierarchy, `HexUnpool` performs no learnable interpolation and does not introduce artifacts or distortions into the surface structure.

Semantic similarity analysis

This section provides a comparative analysis of the biomedical language models used in the experiments, with the goal of motivating their selection and highlighting differences in semantic behaviour that are relevant for anatomical text prompts.

Table 1 reports cosine similarities between the term "frontal lobe" and several related or distinct brain-region terms across multiple biomedical language models. Models pretrained on large biomedical or radiological corpora (e.g., BioClinicalBERT, PubMedBERT, RadBERT-RoBERTa) exhibit more consistent similarity patterns across semantically related regions, whereas models trained on general-domain text show less anatomically meaningful behaviour.

Table 1: Cosine similarity between "frontal lobe" and related region terms across different biomedical language models.

Model	Prefrontal cortex	Inferior frontal gyrus	Right	Right frontal	Temporal lobe	Parietal lobe
bionlp/bluebert_pubmed_mimic_uncased_L-12_H-768_A-12	0.882	0.812	0.792	0.904	0.968	0.892
emilyalsentzer/Bio_ClinicalBERT	0.909	0.877	0.781	0.941	0.982	0.932
microsoft/BiomedNLP-PubMedBERT-base-uncased-abstract-fulltext	0.974	0.972	0.931	0.971	0.985	0.987
StanfordAIMI/RadBERT	0.504	0.431	0.369	0.654	0.816	0.599
zzxslp/RadBERT-RoBERTa-4m	0.971	0.946	0.921	0.960	0.971	0.976
microsoft/BiomedVLP-CXR-BERT-general	0.489	0.623	0.572	0.680	0.770	0.810
cambridgeltl/SapBERT-from-PubMedBERT-fulltext	0.762	0.587	0.360	0.689	0.577	0.703
allenai/scibert_scivocab_uncased	0.903	0.923	0.748	0.917	0.946	0.976
intfloat/e5-base	0.893	0.899	0.771	0.892	0.892	0.889

When comparing regions that are subsets of the frontal lobe ("prefrontal cortex", "inferior frontal gyrus") with lateral or orientation-only terms ("right", "right frontal"), we observe the following pattern:

- frontal-lobe subsets show high similarity to "frontal lobe" (average 0.81 for "prefrontal cortex" and 0.79 for "inferior frontal gyrus");
- the direction-only term "right" is consistently lower (around 0.69);
- the compound term "right frontal"—which shares the lexical component "frontal"—shows higher similarity (around 0.85).

These results indicate that biomedical language models primarily capture lexical overlap and contextual co-occurrence rather than true anatomical relationships. Terms that share components or frequently appear in similar scientific contexts tend to form closely clustered embeddings, even when they refer to distinct anatomical regions. For example, any expression containing "frontal" is placed near "frontal lobe" due to shared vocabulary and usage patterns. Similarly, other major cortical lobes (e.g., temporal, parietal) also show relatively high similarity to "frontal lobe", reflecting a tendency of biomedical transformers to group broadly defined cortical structures. This behaviour is typical across models and should not be interpreted as anatomical grounding.

When selecting a model for our experiments, our aim is not to recover the underlying anatomical hierarchy but to identify a model that handles domain terminology in a stable and predictable manner. In particular, we prioritise models that capture semantic relatedness between terms that occur in similar linguistic contexts and that distinguish compound expressions from more general ones. Such stability in semantic patterns is sufficient for our downstream pipeline.

Based on these observations, **PubMedBERT** and **BlueBERT** demonstrate the most consistent and task-relevant behaviour and are therefore suitable for generating text embeddings in our framework. In contrast, **StanfordAIMI/RadBERT**, **BiomedVLP-CXR-BERT-general**, and **SapBERT** exhibit lower or less stable similarities among related terms and are consequently less suitable for our purposes.

Text distribution

To characterise potential sources of bias, we analyse the distribution of text-derived region labels at three anatomical levels: hemispheres, lobes, and lobe-regions. This reveals whether particular regions or hemispheres are overrepresented, which could drive overfitting or induce a preference for frequent anatomical terms during training.

Unless stated otherwise, all statistics are computed on the *entire* dataset (train, validation, and test). The label *No lesion detected* denotes healthy controls. These cases are excluded from all lesion-based evaluation metrics, but are retained in the test set exclusively for computing specificity. As a result, the dominance of the *No lesion detected* label is reduced in lesion–based analyses, although class imbalance across anatomical regions remains.

Hemisphere. The distribution of hemisphere labels is shown in Figure 1. In the table *"No lesion detected"* means these are healthy patients.

In the tables, *"No lesion detected"* denotes **healthy control** scans with no radiologically/annotationally confirmed FCD.

The number of cases for the left and right hemispheres is approximately equal, with only a small, statistically insignificant difference. This indicates that the *entire dataset* is well balanced across hemispheres. Consequently, a model trained on these data is unlikely to learn a systematic bias towards either hemisphere, which would otherwise reduce its ability to generalise.

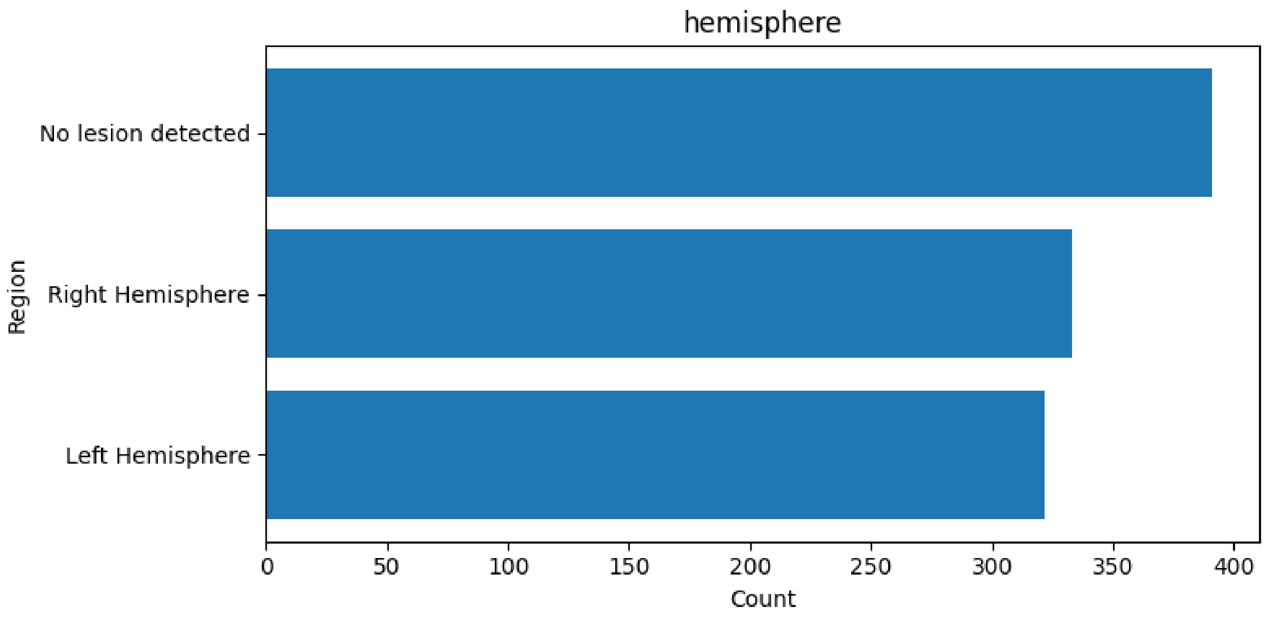


Figure 1: Hemisphere distribution

Lobes. The lobe-level distribution, presented in Figure 2, demonstrates a moderate imbalance. The *frontal*, *limbic*, and *parietal* lobes occur much more frequently than the *occipital*, *insular*, *subcallosal*, and especially the *brainstem*, which are relatively rare. Such imbalance may cause the model to "memorise" common patterns such as *frontal lobe* while underperforming on underrepresented categories like *brainstem* or *insula*. However, compared to the following plot, the lobe regons imbalance can be considered moderate and still suitable for training, especially if loss weighting or other balancing techniques are used.

Lobe regions. The third plot (Figure 3) provides the distribution for specific anatomical

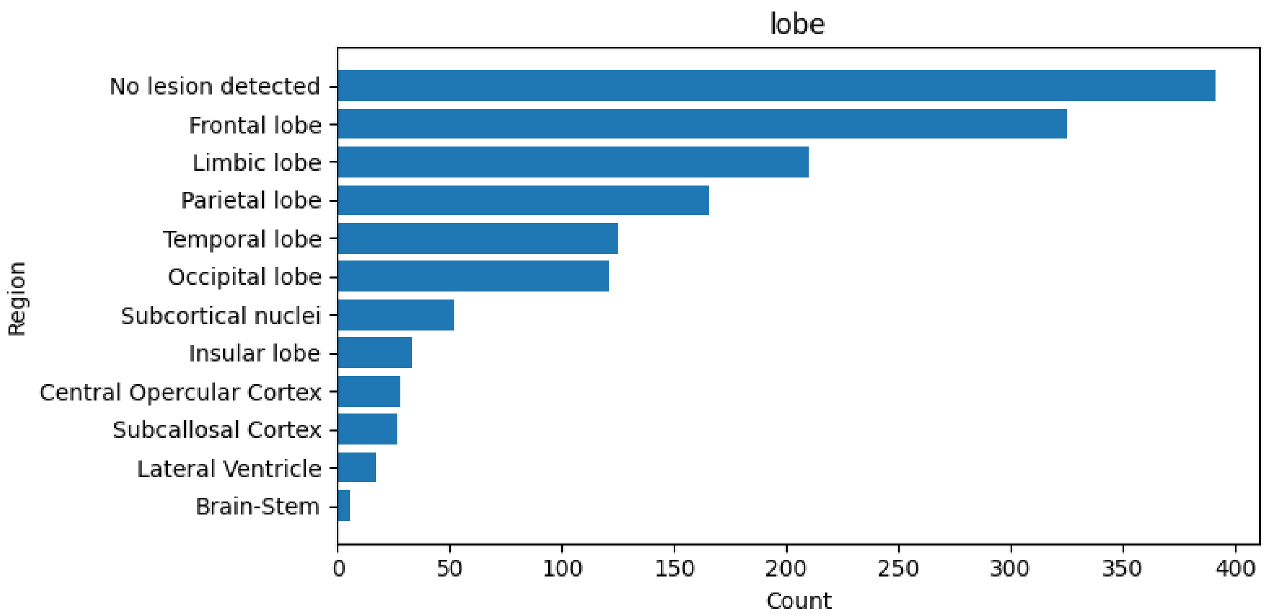


Figure 2: Lobe distribution

regions within the lobes. Here, the imbalance becomes much more pronounced. The label *No lesion detected* dominates the dataset, with a frequency exceeding 400, whereas many regions appear only one to three times. This extremely long-tailed distribution suggests a risk of overfitting to the most frequent labels. As a result, the model may bias its *spatial* predictions toward regions whose textual labels are frequent in the training data (e.g., the *frontal gyrus*), assigning higher lesion probabilities there. This imbalance poses a risk for models relying on textual embeddings, since they may learn frequency-driven rather than semantically meaningful representations.

Discussion. From these findings, we conclude that using fine-grained region names directly as text input would likely lead to model overfitting, given the strong imbalance and the large number of rare categories. The extreme class skew among fine-grained lobe–region labels makes direct use of their verbatim names as text inputs ill-suited: models tend to memorise frequent terms and underfit rare ones, which harms generalisation. To further mitigate imbalance, one could consider data augmentation, class-weighted loss functions, or over-sampling subjects with lesions in rare anatomical regions, at the cost of longer training times and increased computational complexity. Nonetheless, such strategies would likely improve model robustness and generalisation performance.

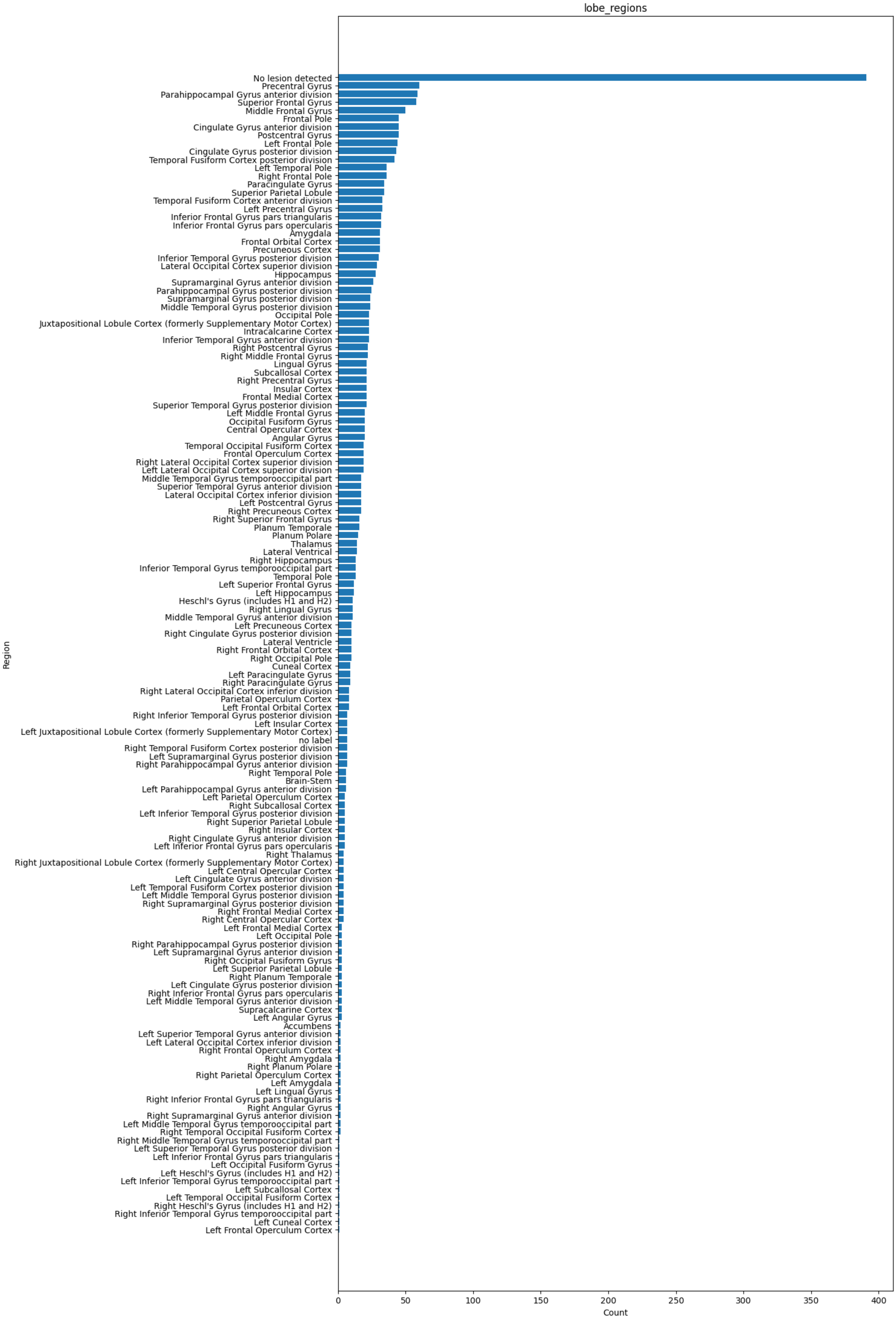


Figure 3: Lobe–region distribution

List of Acronyms

CT Computed Tomography

EEG Electroencephalography

FCD Focal Cortical Dysplasia

GNN Graph Neural Network

IoU Intersection over Union

MELD Multicentre Epilepsy Lesion Detection

MRI Magnetic Resonance Imaging

p.p. percentage points

PPV Positive Predictive Value

ROI Region of Interest

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